Compal Confidential

KALH0 /KAL90+ /KALG0 M/B Schematics Document

Intel Penryn Processor with Cantiga + DDRIII + ICH9M

2009-3-4

REV:1.0

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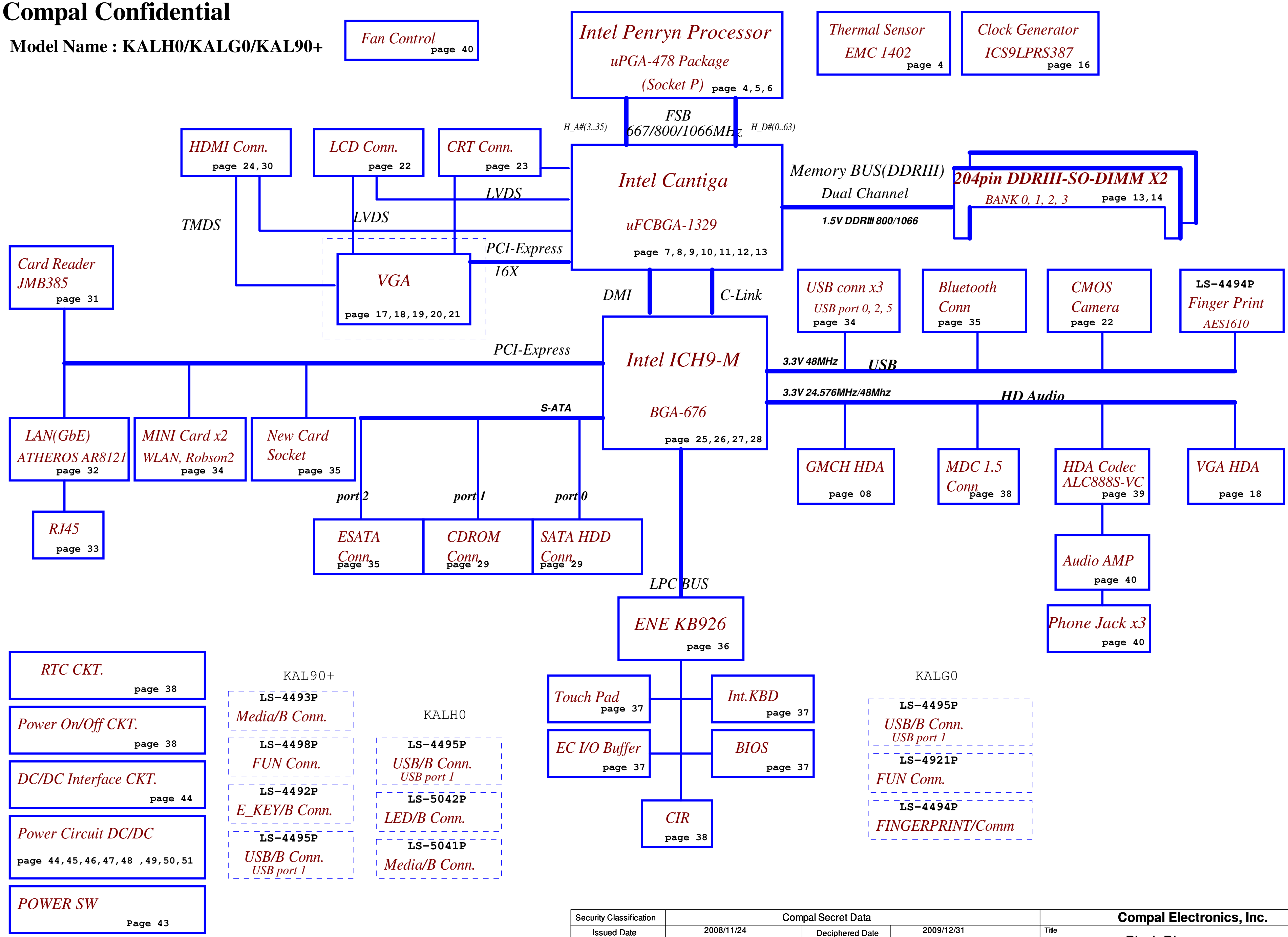
Model Name : KALH0/KALG0/KAL90+



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STATE \ SIGNAL	SLP_S1#	SLP_S3#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON	HIGH	HIGH	HIGH	HIGH	ON	ON	ON	ON
S1(Power On Suspend)	LOW	HIGH	HIGH	HIGH	ON	ON	ON	LOW
S3 (Suspend to RAM)	LOW	LOW	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)	LOW	LOW	LOW	HIGH	ON	OFF	OFF	OFF
S5 (Soft OFF)	LOW	LOW	LOW	LOW	ON	OFF	OFF	OFF

Voltage Rails

Power Plane	Description	S1	S3	S5
VIN	Adapter power supply (19V)	N/A	N/A	N/A
B+	AC or battery power rail for power circuit.	N/A	N/A	N/A
+CPU_CORE	Core voltage for CPU	ON	OFF	OFF
+0.75VS	0.75V switched power rail for DDR terminator	ON	OFF	OFF
+1.05VS	1.05V switched power rail	ON	OFF	OFF
+1.25VS	1.25V switched power rail	ON	OFF	OFF
+1.5V	1.5V power rail for HDA/DDR3	ON	ON	OFF
+1.5VS	1.5V switched power rail	ON	OFF	OFF
+1.8V	1.8V GM LVDS MODULE	ON	ON	OFF
+1.8VS	1.8V switched power rail	ON	OFF	OFF
+1.1VS	1.1V switched power rail	ON	OFF	OFF
+3VALW	3.3V always on power rail	ON	ON	ON*
+3V	3.3V power rail for SB	ON	ON	X
+3V_LAN	3.3V power rail for LAN	ON	ON	X
+3VS	3.3V switched power rail	ON	OFF	OFF
+5VALW	5V always on power rail	ON	ON	ON*
+5VS	5V switched power rail	ON	OFF	OFF
+VSB	VSB always on power rail	ON	ON	ON*
+RTCVCC	RTC power	ON	ON	ON
+VGA_CORE	Core voltage for GPU	ON	OFF	OFF

Note : ON* means that this power plane is ON only with AC power available, otherwise it is OFF.

Board ID / SKU ID Table for AD channel

Vcc	3.3V +/- 5%
Ra/Rc/Re	100K +/- 5%

Board ID	Rb / Rd / Rf	V_{AD_BID} min	V_{AD_BID} typ	V_{AD_BID} max
0	0	0 V	0 V	0 V
1	8.2K +/- 5%	0.216 V	0.250 V	0.289 V
2	18K +/- 5%	0.436 V	0.503 V	0.538 V
3	33K +/- 5%	0.712 V	0.819 V	0.875 V
4	56K +/- 5%	1.036 V	1.185 V	1.264 V
5	100K +/- 5%	1.453 V	1.650 V	1.759 V
6	200K +/- 5%	1.935 V	2.200 V	2.341 V
7	NC	2.500 V	3.300 V	3.300 V

BOARD ID Table

Board ID	PCB Revision
0	0.1
1	0.2
2	0.3
3	1.0
4	1A
5	
6	
7	

BTO Option Table

BTO Item	BOM Structure
KAL90	KAL90@
UMA	GM@
PM@	PM@
ALC888VC	888VC@
ALC888VB	888VB@
AR8121	8121@
AR8112	8112@
ALC268	268@
GL40	GL40@
GM45	GM45@
KAL90-G0	KAL90_G0@
KAL90-H0	KAL90_H0@
KALG0	KALG0@
KALH0	KALH0@
ALC268	268@
	KAL90_90+@
	KAL90_H0_G0@
	KAL90+_G0
	KALH0_G0
	KAL90_G0_90+@
	KAL90_H0_90+@
	KAL90+_PCB@
	KALG0_PCB@

External PCI Devices

Device	IDSEL#	REQ#/GNT#	Interrupts

EC SM Bus1 address

Device	Address
Smart Battery	0001 011X b
MEDIA CONSOLE	1010 000X b

EC SM Bus2 address

Device	Address
ADI ADT7421	1001 100X b
NB9M THERMAL SENSOR	

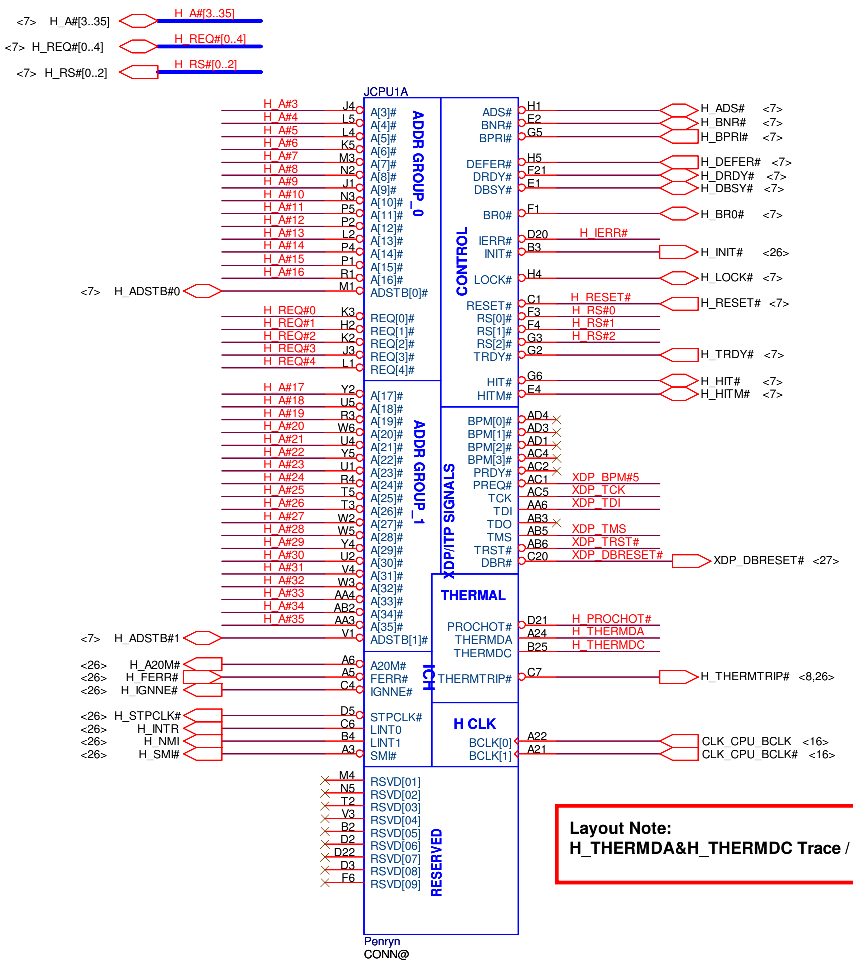
ICH9M SM Bus address

Device	Address
Clock Generator (ICS9LPRS387, SLG8SP556V)	1101 001Xb
DDR DIMM0	1001 000Xb
DDR DIMM2	1001 010Xb

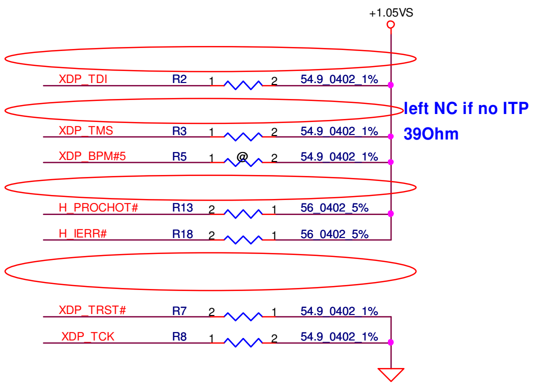
BOM Configuration Table

Project	BOM Configuration
KAL90-UMA	XXXXXXXXXX:KAL90@/GM@/888VC@/8121@/GM45@
KAL90-Dis	XXXXXXXXXX:KAL90@/PM@/888VC@/8121@
KALH0-GM45	XXXXXXXXXX:KALH0@/GM@/888VC@/8121@/GM45@
KALH0-GL40	XXXXXXXXXX:KALH0@/GM@/888VC@/8121@/GL40@
KALH0-PM45	XXXXXXXXXX:KALH0@/PM@/888VC@/8121@
KAL90+ -UMA	GM@/888VC@/8121@/GM45@/KAL90+_G0@/KAL90_90+@/KAL90_G0_90+@/KAL90_H0_90+@/KAL90+_PCB@
KAL90+ -Dis	PM@/888VC@/8121@/KAL90+_G0@/KAL90_90+@/KAL90_G0_90+@/KAL90_H0_90+@/KAL90+_PCB@/PM45@
KALG0 -UMA(GL40)	KALG0@/GM@/888VC@/8121@/GL40@/KAL90+_G0@/KALH0_G0@/KAL90_G0_90+@/KALG0_DDR2 PCB RV0 @/KALG0+@
KALG0 -Dis	KALG0@/PM@/888VC@/8121@/PM45@/KAL90+_G0@/KALH0_G0@/KAL90_G0_90+@/KALG0_DDR2 PCB RV0 @/KALG0+@
KALG0 -UMA(GM45)	KALG0@/GM@/888VC@/8121@/GM45@/KAL90+_G0@/KALH0_G0@/KAL90_G0_90+@/KALG0_DDR2 PCB RV0 @/KALG0+@
KALG0 -DIS(GM45)	KALG0@/PM@/888VC@/8121@/GM45@/KAL90+_G0@/KALH0_G0@/KAL90_G0_90+@/KALG0_DDR2 PCB RV0 @/KALG0+@

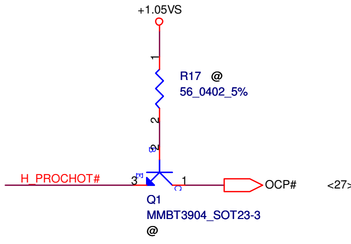
KALG0 LAN to AR-8131------- 8121@ Change to 8131@

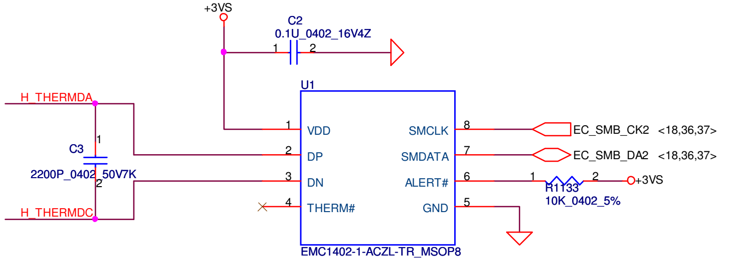


Layout Note:
H_THERMDA&H_THERMDC Trace / Space = 10 / 10 mil



BSEL2	BSEL1	BSEL0	BCLK
0	0	0	266
0	1	0	200
0	1	1	166





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Size: B | Document Number: KALH0/KALG0/KAL90+ | Rev: 1.0

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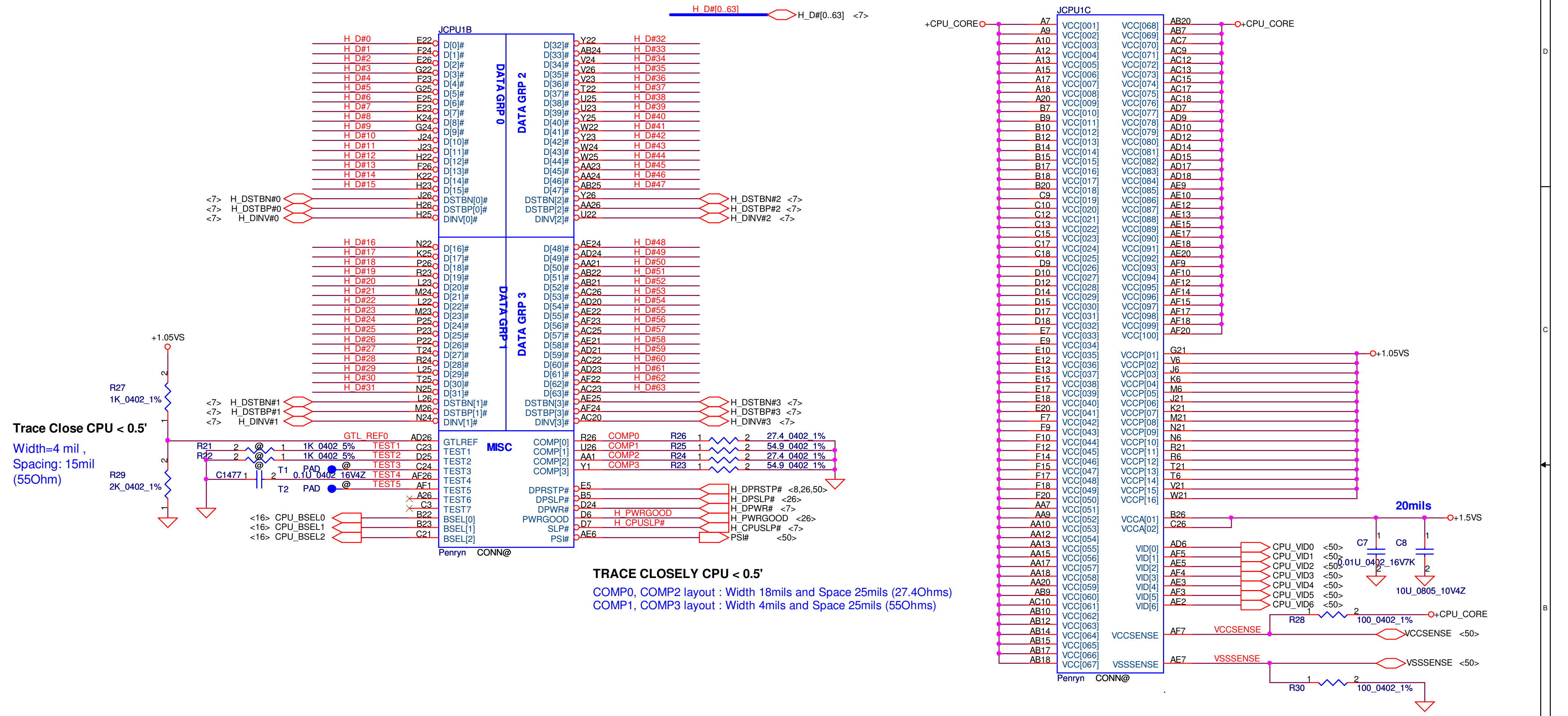

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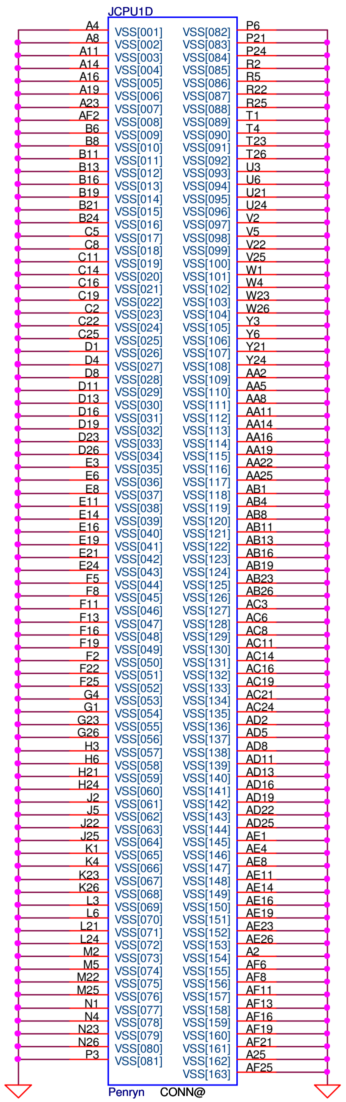
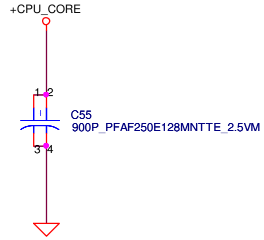
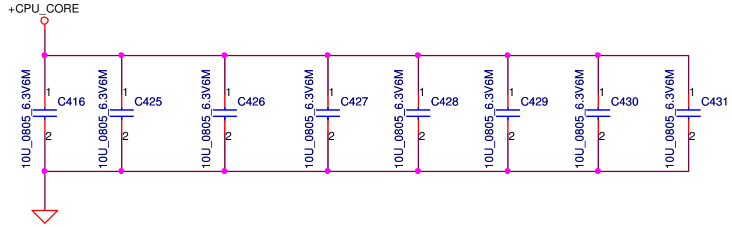
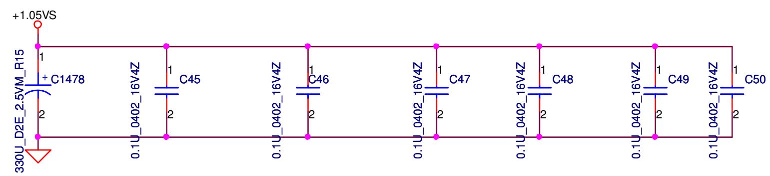






+CPU-CORE Decoupling	C,uF	ESR, mohm	ESL,nH
SPCAP,Polymer	4X330uF	6m ohm/4	1.8nH/6
MLCC 0805 X5R	32X22uF	3m ohm/32	0.6nH/32
	32X10uF	3m ohm/32	0.6nH/32



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Title: Penryn (3/3)

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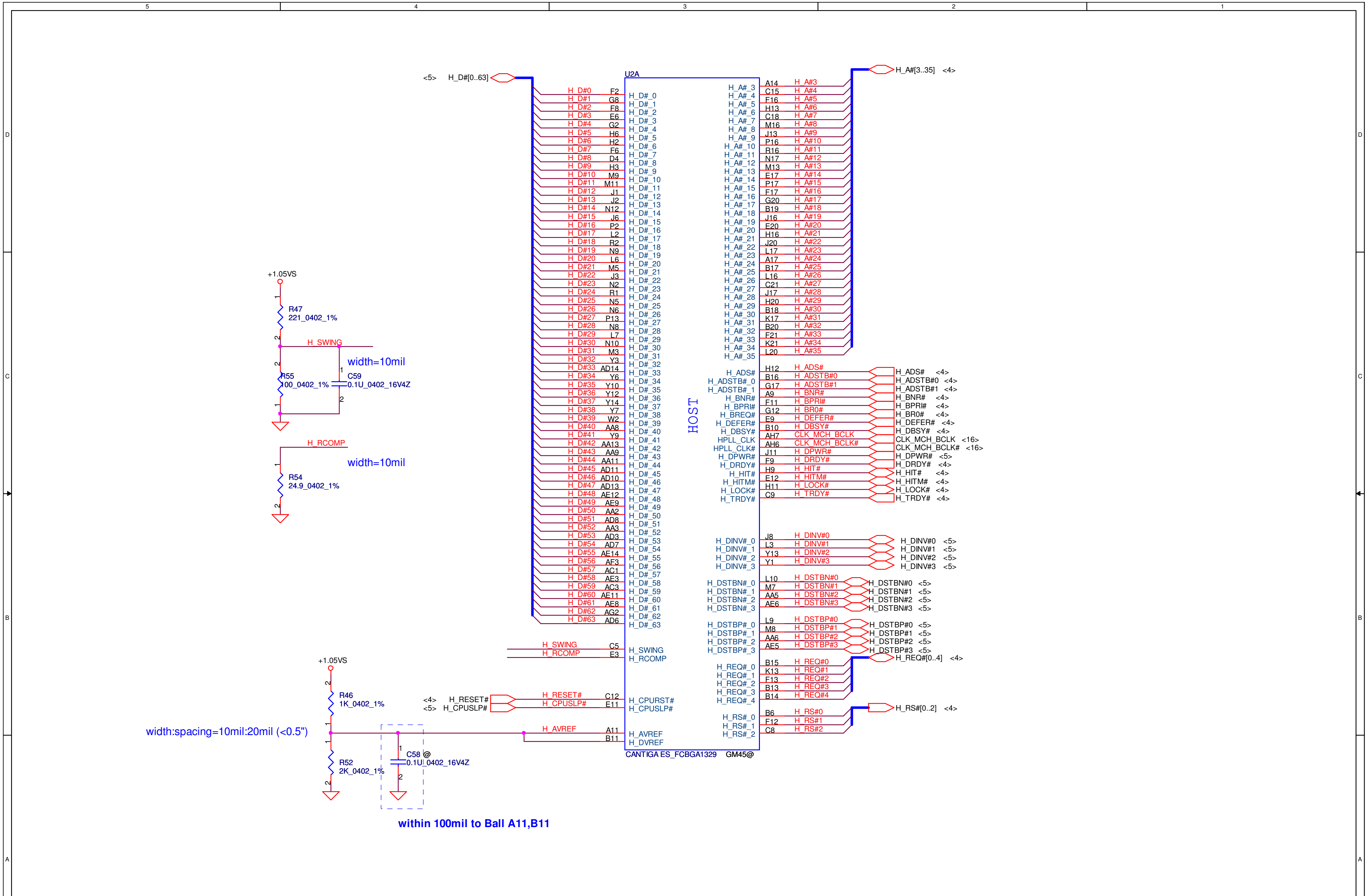

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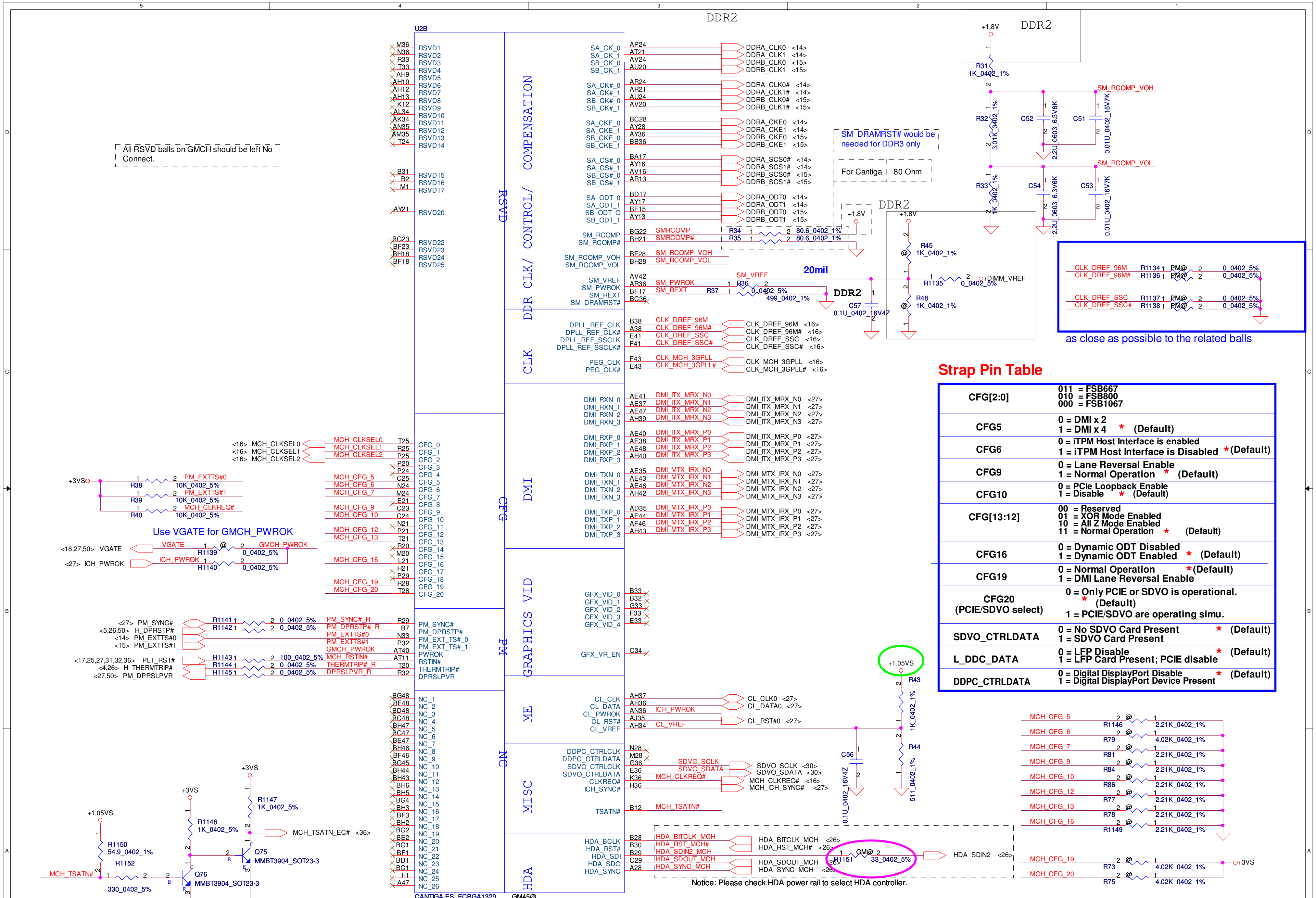
Compal Electronics, Inc.

Title: Cantiga GMCH(1/7)-GTL

Size: B | Document Number: KALH0/KALG0/KAL90+ | Rev: 1.0

Date: Monday, April 27, 2009 |



Strap Pin Table

CFG[2:0]	011 = FSB667 010 = FSB800 000 = FSB1067
CFG5	0 = DMI x 2 1 = DMI x 4 * (Default)
CFG6	0 = iTPM Host Interface is enabled 1 = iTPM Host Interface is Disabled *(Default)
CFG9	0 = Lane Reversal Enable 1 = Normal Operation * (Default)
CFG10	0 = PCIe Loopback Enable 1 = Disable * (Default)
CFG[13:12]	00 = Reserved 01 = XOR Mode Enabled 10 = All Z Mode Enabled 11 = Normal Operation * (Default)
CFG16	0 = Dynamic ODT Disabled 1 = Dynamic ODT Enabled * (Default)
CFG19	0 = Normal Operation *(Default) 1 = DMI Lane Reversal Enable
CFG20 (PCIE/SDVO select)	0 = Only PCIE or SDVO is operational. * (Default) 1 = PCIE/SDVO are operating simu.
SDVO_CTRLDATA	0 = No SDVO Card Present * (Default) 1 = SDVO Card Present
L_DDC_DATA	0 = LFP Disable * (Default) 1 = LFP Card Present; PCIE disable
DDPC_CTRLDATA	0 = Digital DisplayPort Disable * (Default) 1 = Digital DisplayPort Device Present

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Compal Electronics, Inc.

Title: Cantiga GMCH(2/7)-DMI/DDR

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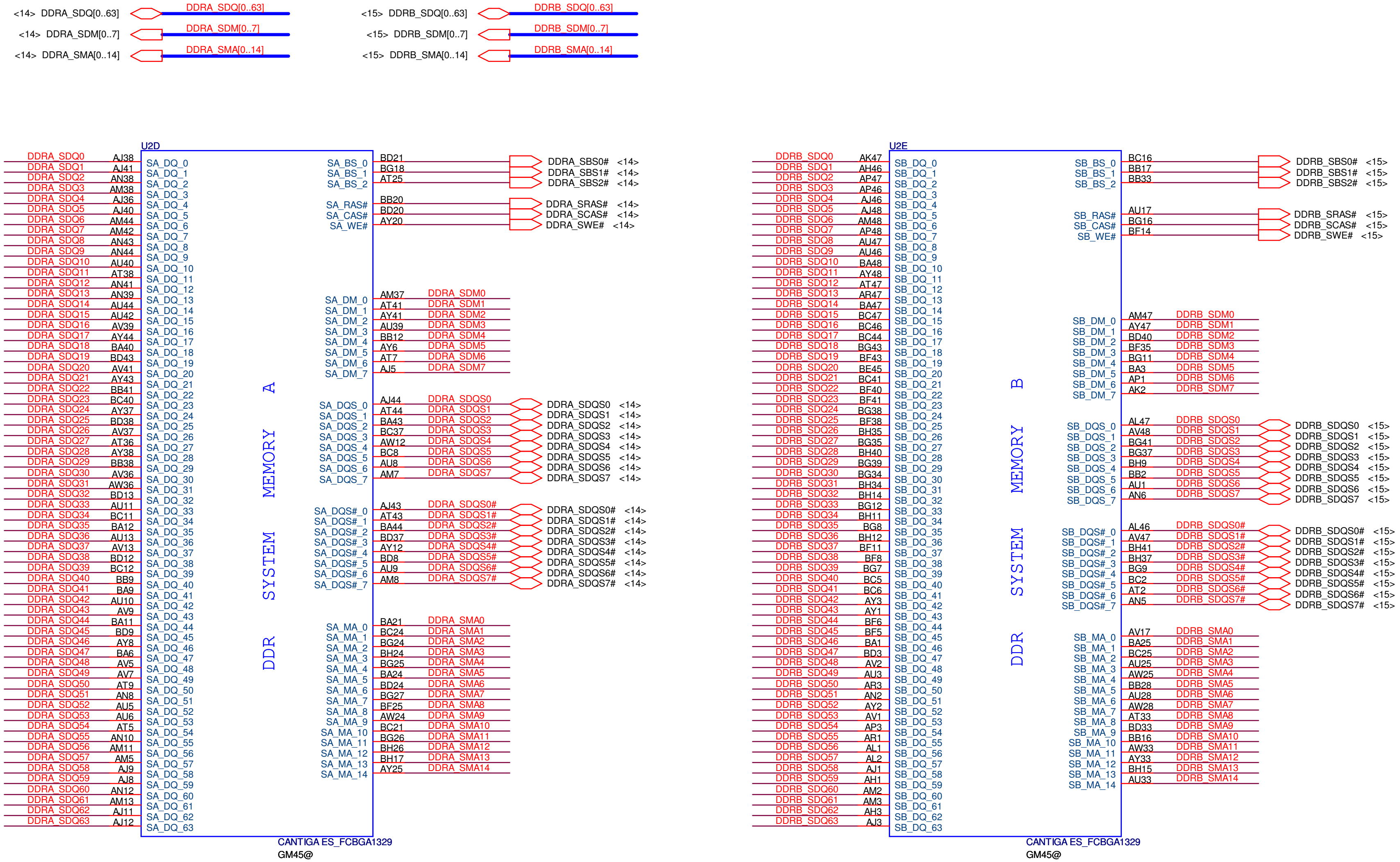

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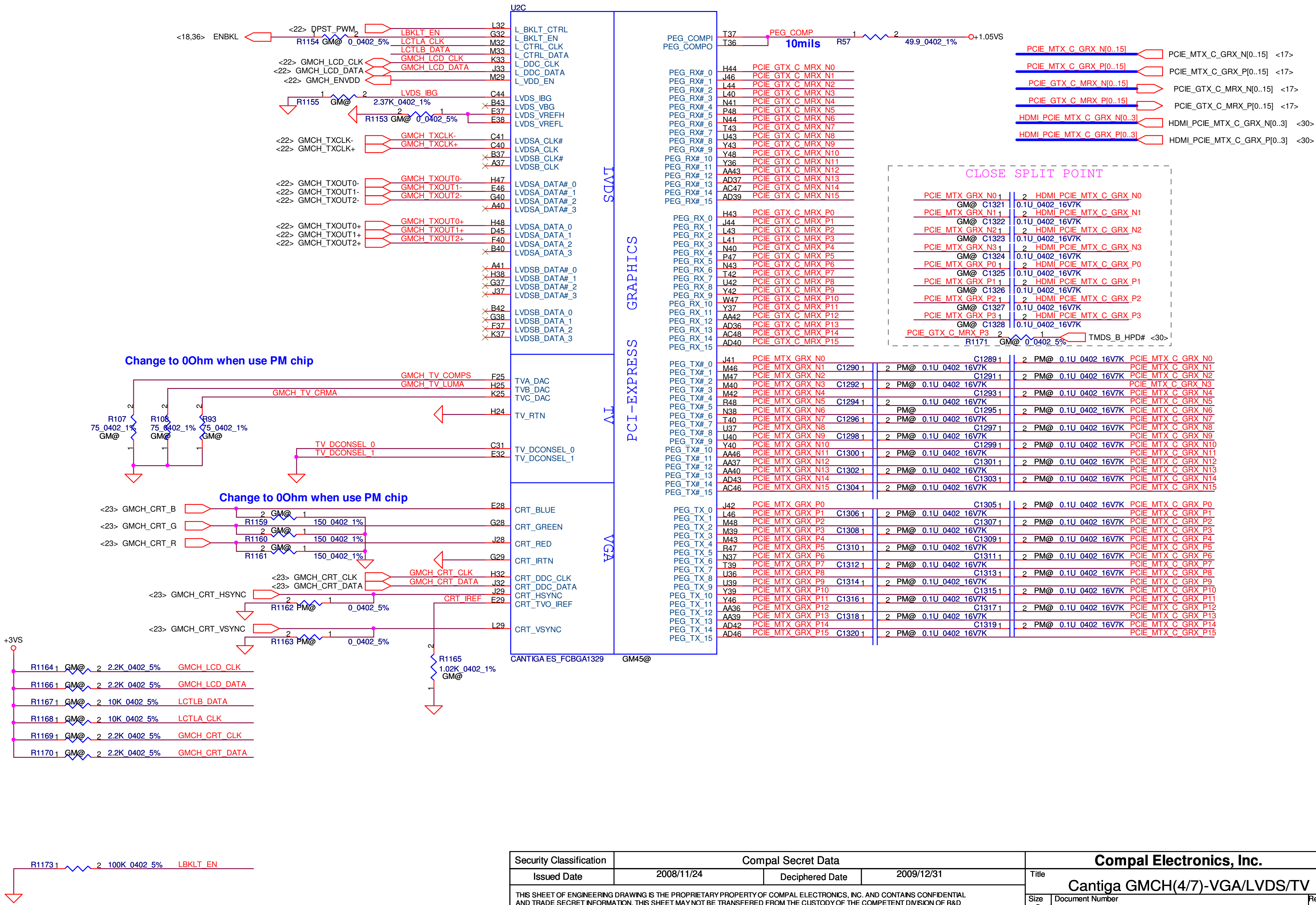
Issued Date	2008/03/28	Deciphered Date	2008/09/20	Title: Cantiga GMCH(3/7)-DDR

Size	Document Number	Rev
B	KAL90	0.1

Date: Monday, April 27, 2009



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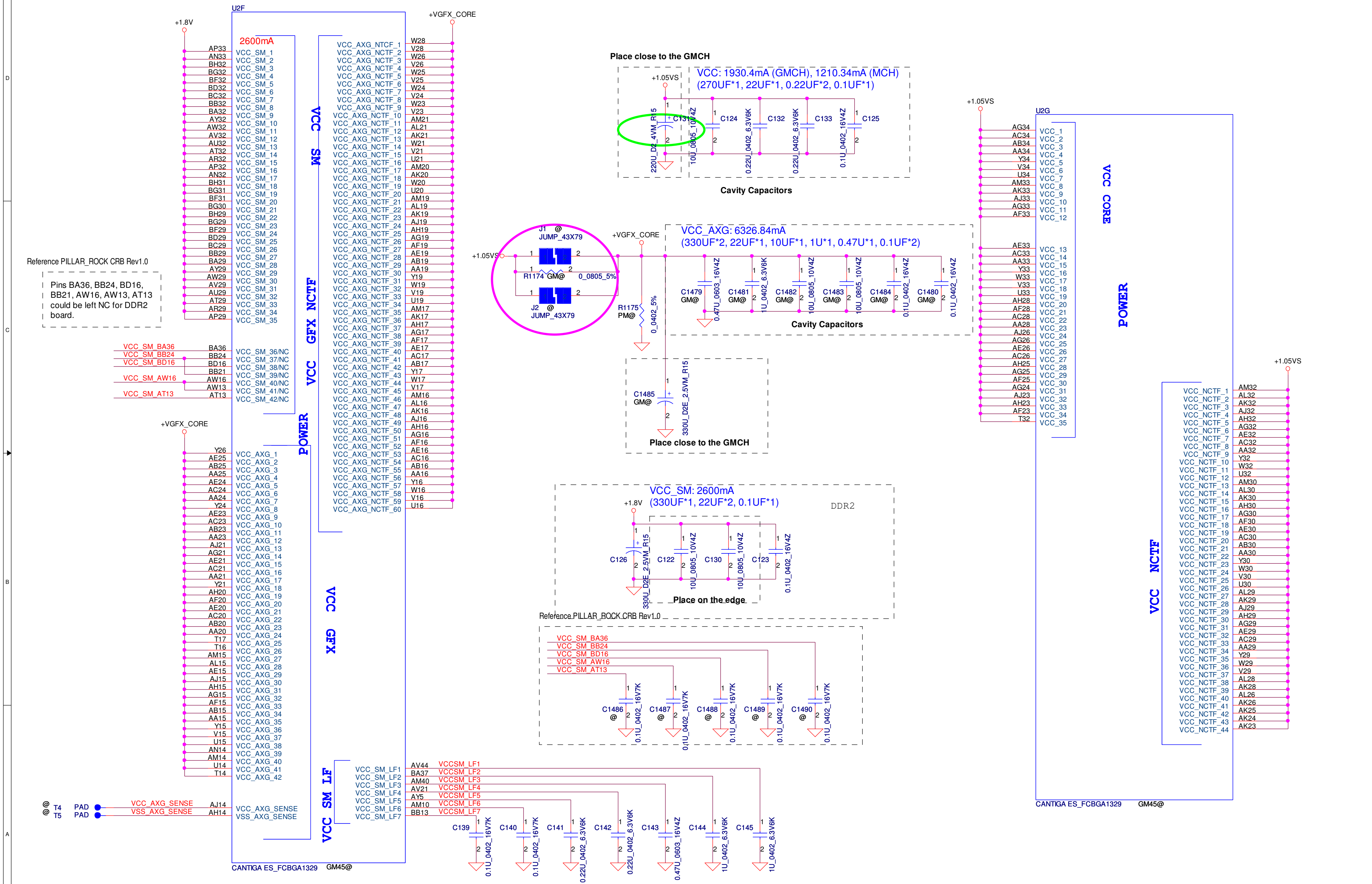

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Title: Cantiga GMCH(5/7)-VCC

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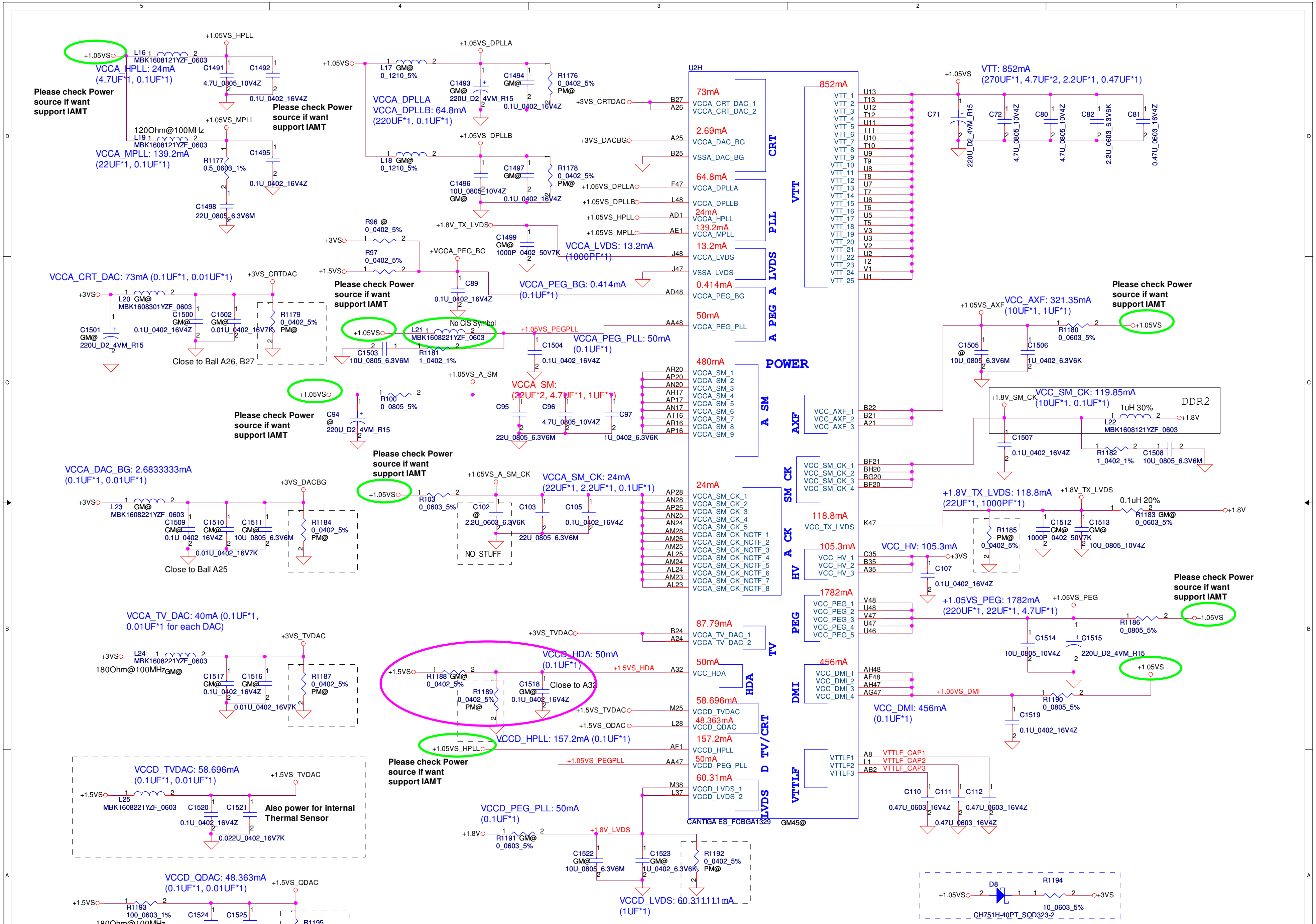

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Title: Crestline GMCH (6/7)-VCC

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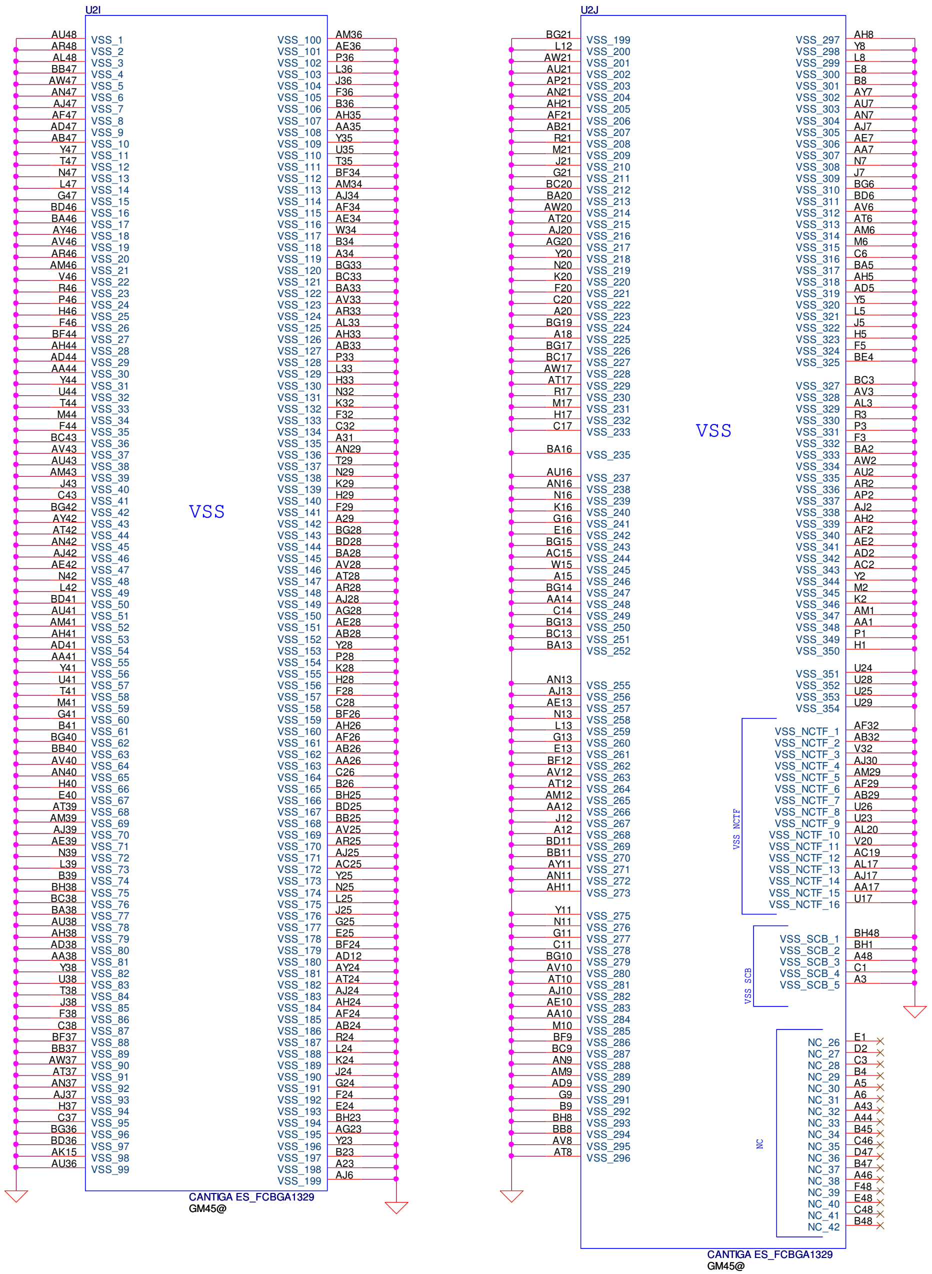

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Compal Electronics, Inc.

Title: Cantiga GMCH(1/7)-GTL

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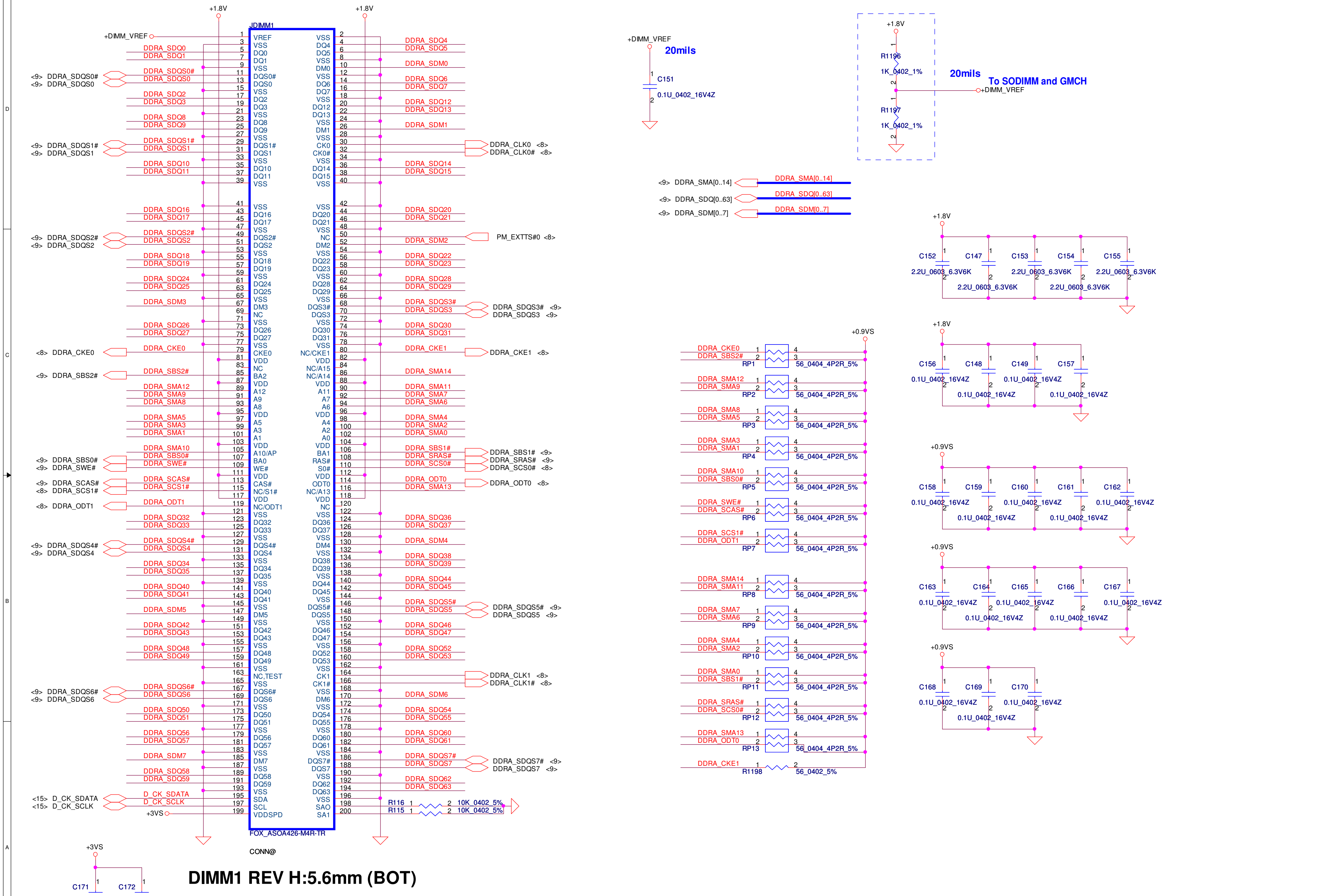

DIMM1 REV H:5.6mm (BOT)

Security Classification	Compal Secret Data		
Issued Date	2008/03/28	Deciphered Date	2008/09/20

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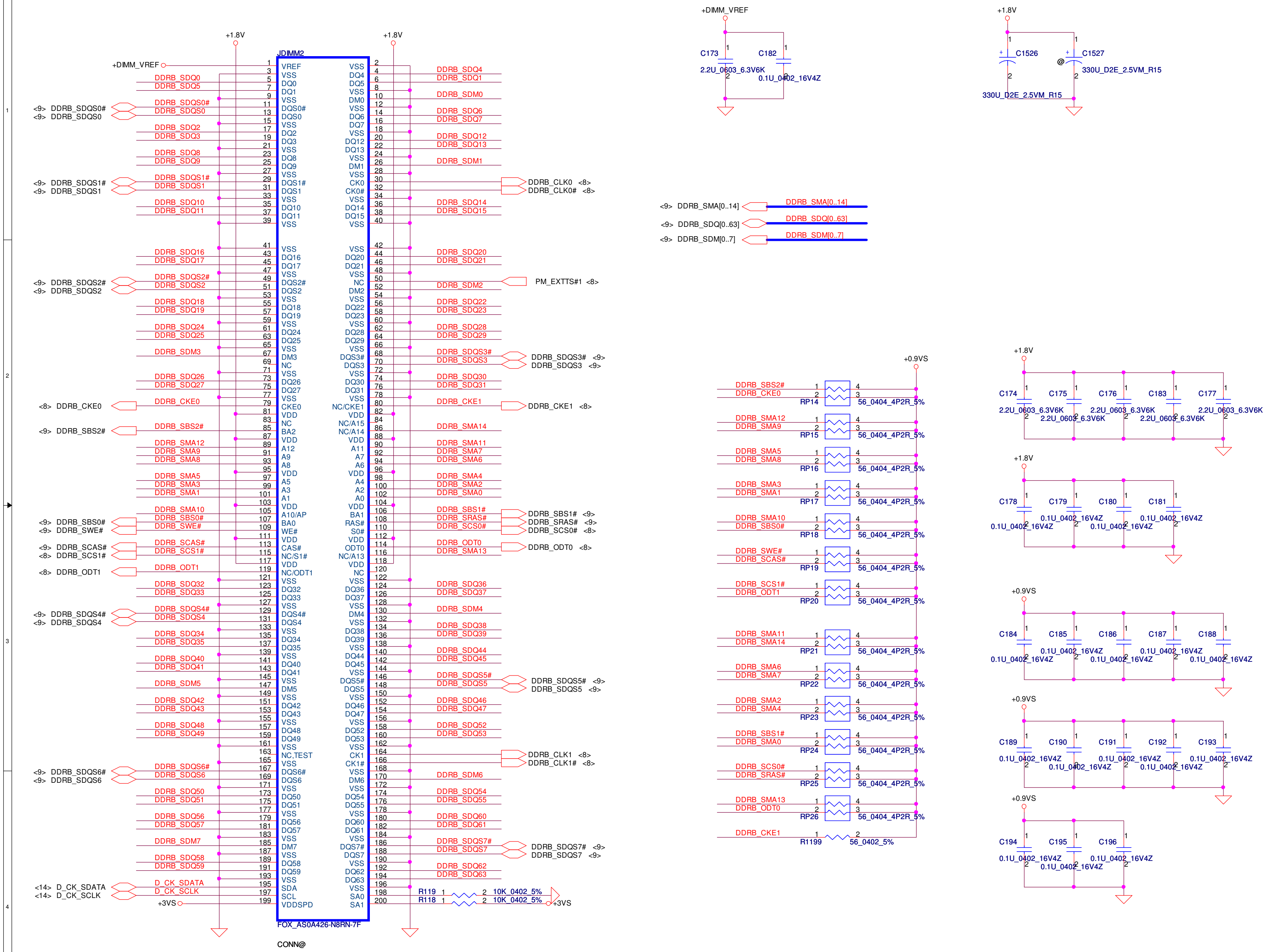
Title: DDRII-SODIMM0

Size: B | Document Number: KAL90 | Rev: 0.1



DIMM2 REV H:10.1mm (BOT)

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Size B | Document Number KAL90 | Rev 0.1

Date: Monday, April 27, 2009 |

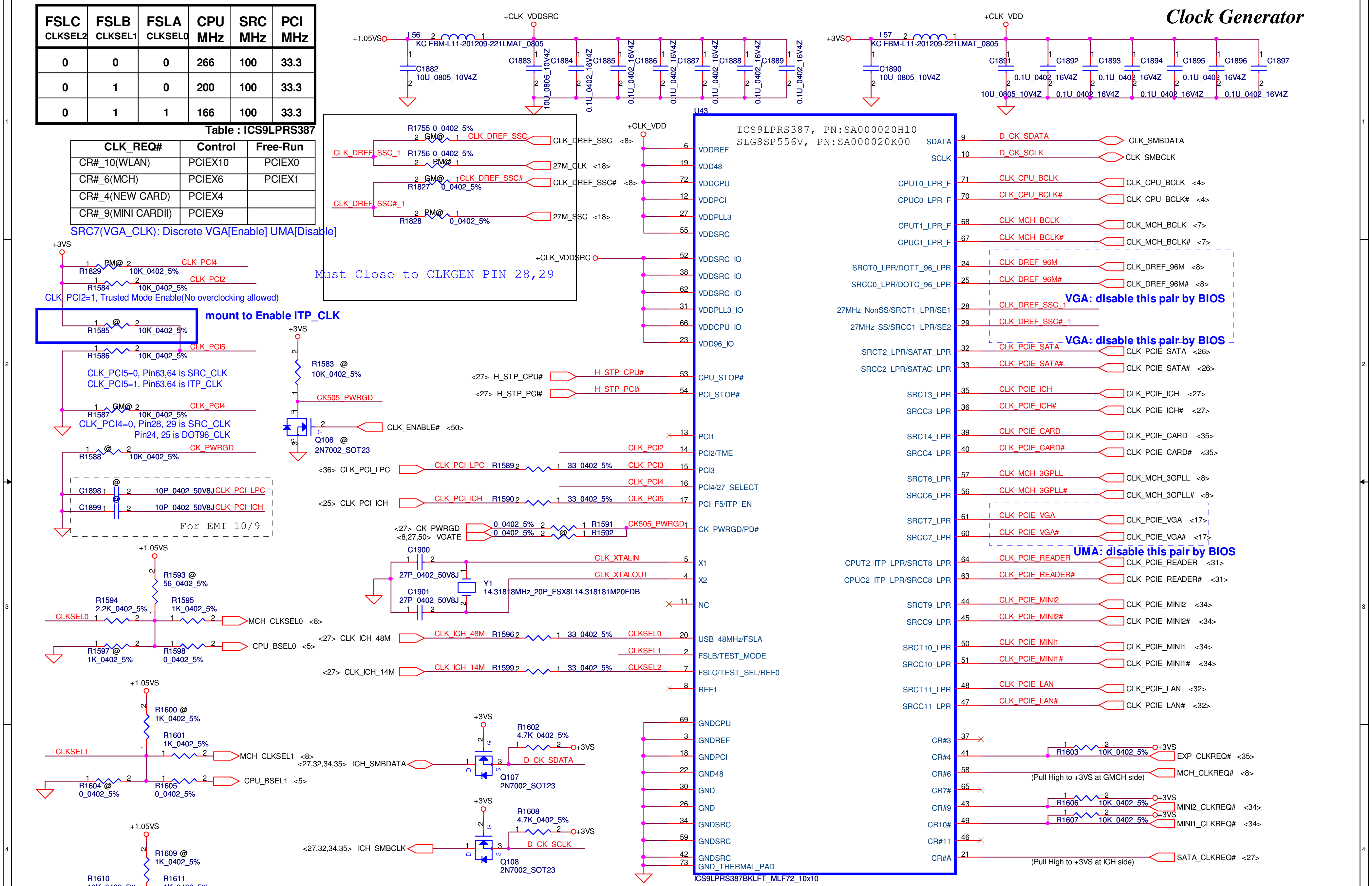
FSLC CLKSEL2	FSLB CLKSEL1	FSLA CLKSEL0	CPU MHz	SRC MHz	PCI MHz
0	0	0	266	100	33.3
0	1	0	200	100	33.3
0	1	1	166	100	33.3

Table : ICS9LPRS387

CLK_REQ#	Control	Free-Run
CR#_10(WLAN)	PCIEX10	PCIEX0
CR#_6(MCH)	PCIEX6	PCIEX1
CR#_4(NEW CARD)	PCIEX4	
CR#_9(MINI CARDII)	PCIEX9	

SRC7(VGA_CLK): Discrete VGA[Enable] UMA[Disable]

Clock Generator



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Title: Clock Generator (CK505)

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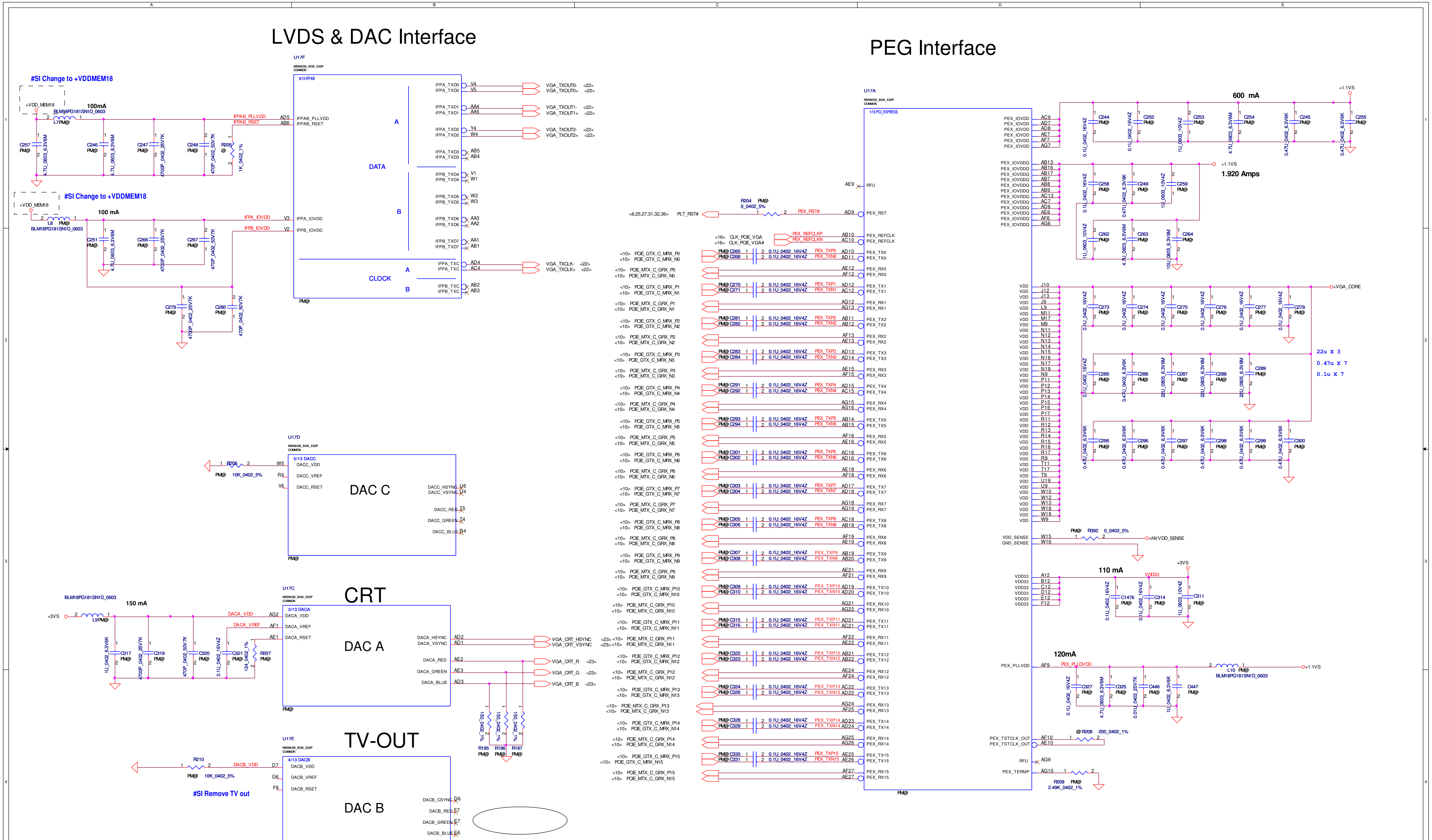

Security Classification	Compal Secret Data		
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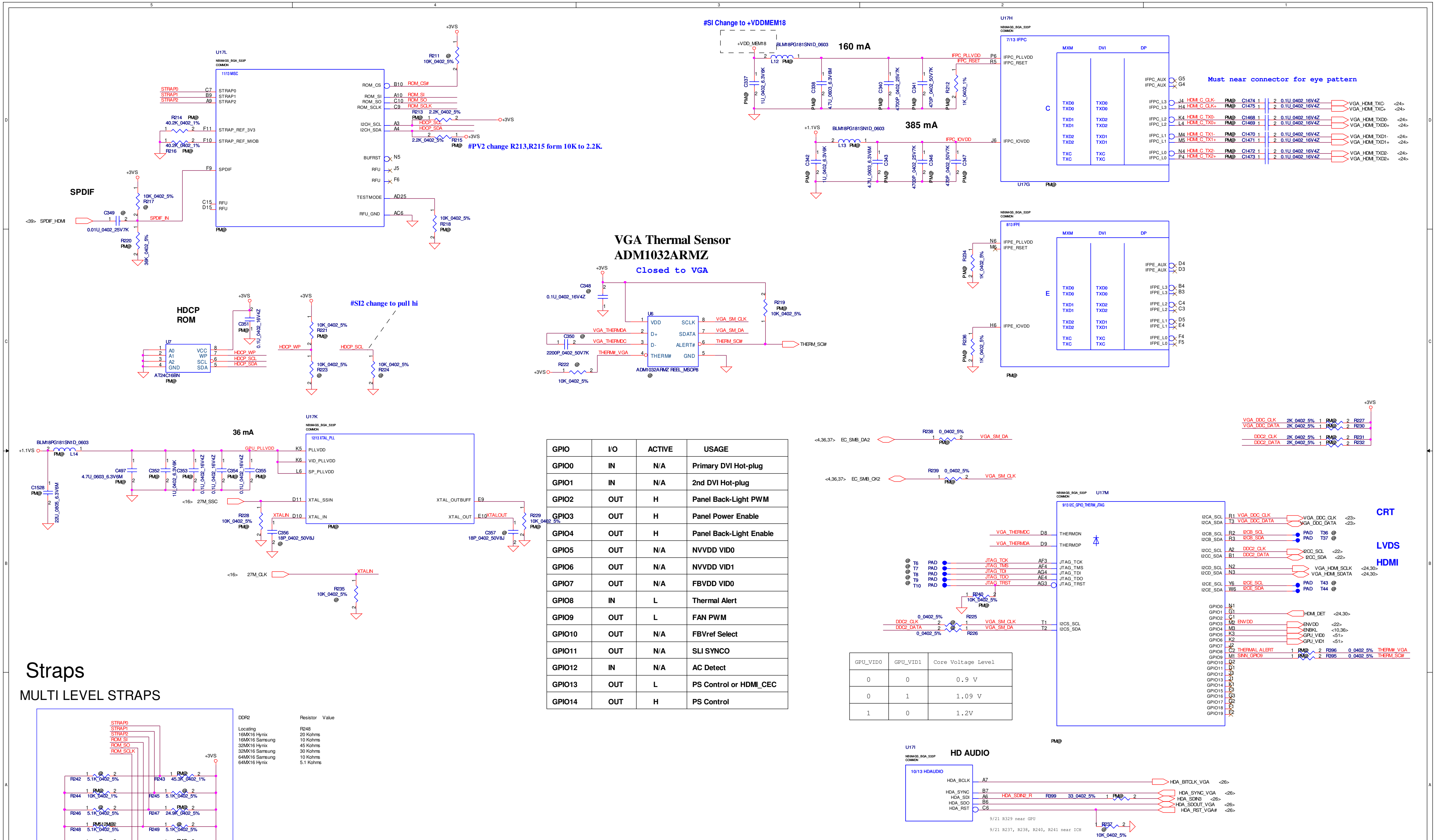
Title: **PEG & LVDS & DAC**

Size	Document Number	Rev
Custom	KALH0/KALG0/KAL90+	1.0

Date: Monday, April 27, 2009 Sheet 17 of 53



GPIO	I/O	ACTIVE	USAGE
GPIO0	IN	N/A	Primary DVI Hot-plug
GPIO1	IN	N/A	2nd DVI Hot-plug
GPIO2	OUT	H	Panel Back-Light PWM
GPIO3	OUT	H	Panel Power Enable
GPIO4	OUT	H	Panel Back-Light Enable
GPIO5	OUT	N/A	NVVDD VID0
GPIO6	OUT	N/A	NVVDD VID1
GPIO7	OUT	N/A	FBVDD VID0
GPIO8	IN	L	Thermal Alert
GPIO9	OUT	L	FAN PWM
GPIO10	OUT	N/A	FBVref Select
GPIO11	OUT	N/A	SLI SYNCO
GPIO12	IN	N/A	AC Detect
GPIO13	OUT	L	PS Control or HDMI_CEC
GPIO14	OUT	H	PS Control

GPU_VID0	GPU_VID1	Core Voltage Level
0	0	0.9 V
0	1	1.09 V
1	0	1.2V

Security Classification	Compal Secret Data	
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Title: **Straps & HDMI**

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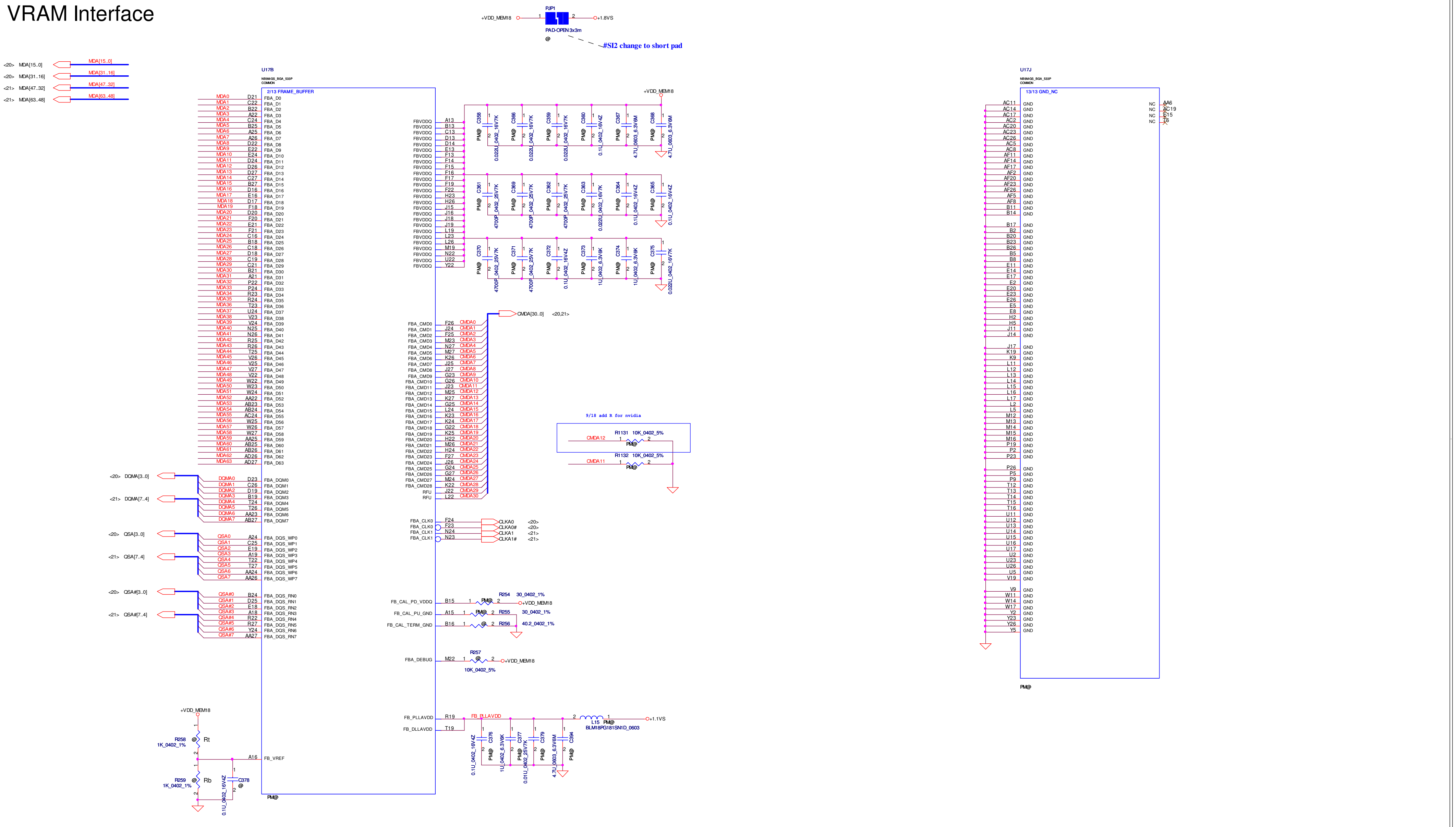
VRAM Interface



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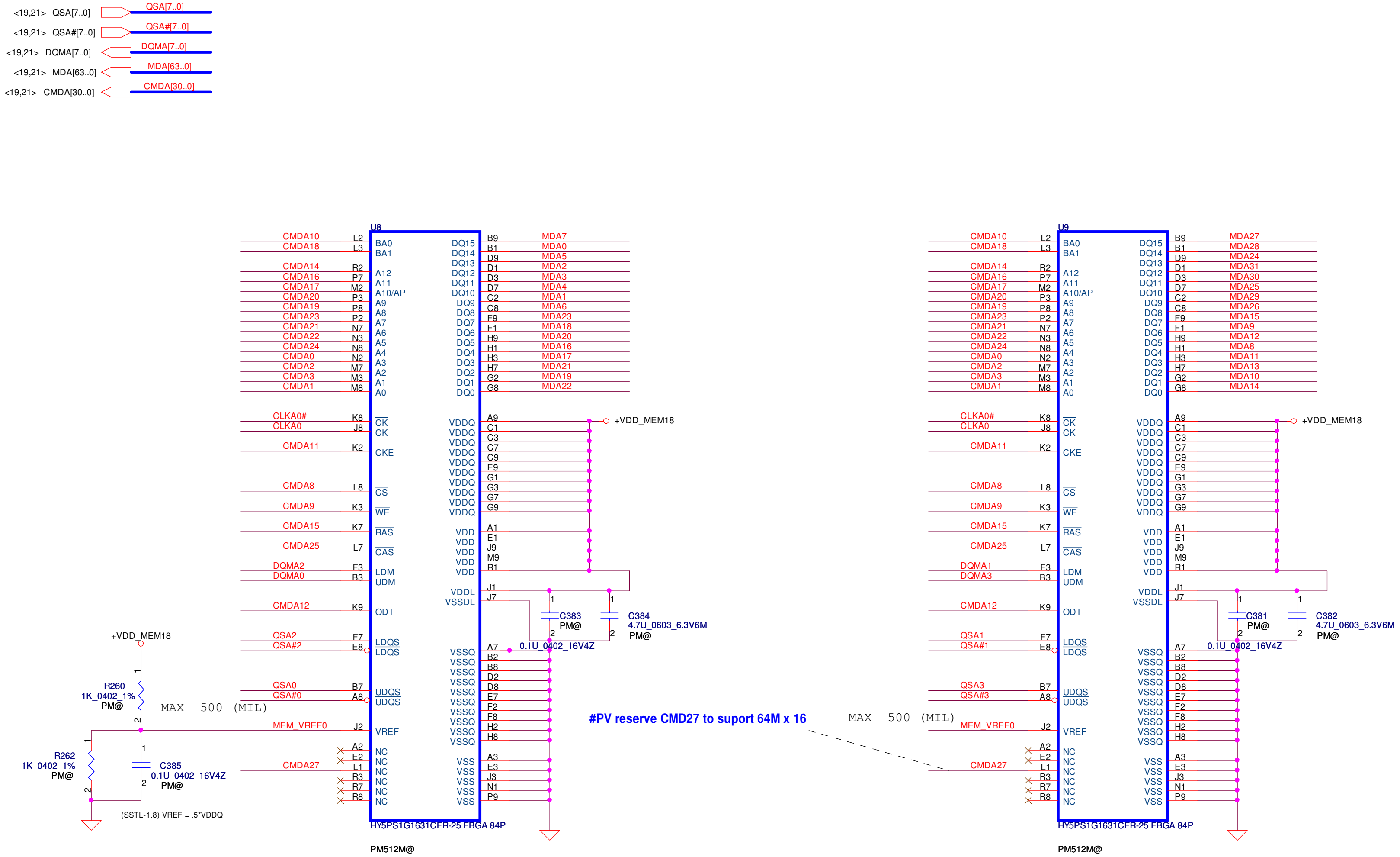
			Date: Monday, April 27, 2009; Sheet 19 of 53

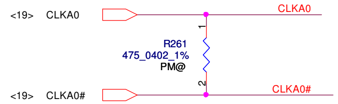
VRAM DDR2 chips (256MB & 512MB)

32Mx16 DDR2 400MHz *4==>256MB

64Mx16 DDR2 400MHz*4==>512MB

	DATA Bus	
Address	0..31	32..63
CMD0	A3	
CMD1	A0	A0
CMD2	A2	
CMD3	A1	A1
CMD4		A3
CMD5		A4
CMD6		A5
CMD7		
CMD8	CS#	CS#
CMD9	WE#	WE#
CMD10	BA0	BA0
CMD11	CKE	CKE
CMD12	ODT	ODT
CMD13		
CMD14	A12	A12
CMD15	RAS#	RAS#
CMD16	A11	A11
CMD17	A10	A10
CMD18	BA1	BA1
CMD19	A8	A8
CMD20	A9	A9
CMD21	A6	A6
CMD22	A5	
CMD23	A7	A7
CMD24	A4	
CMD25	CAS#	CAS#
CMD26	A13	A13
CMD27	BA2	BA2
CMD28		
CMD29		
CMD30		





Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

Compal Electronics, Inc.

Title: VRAM 1

Size | Document Number: KAL90 | Rev 0.1

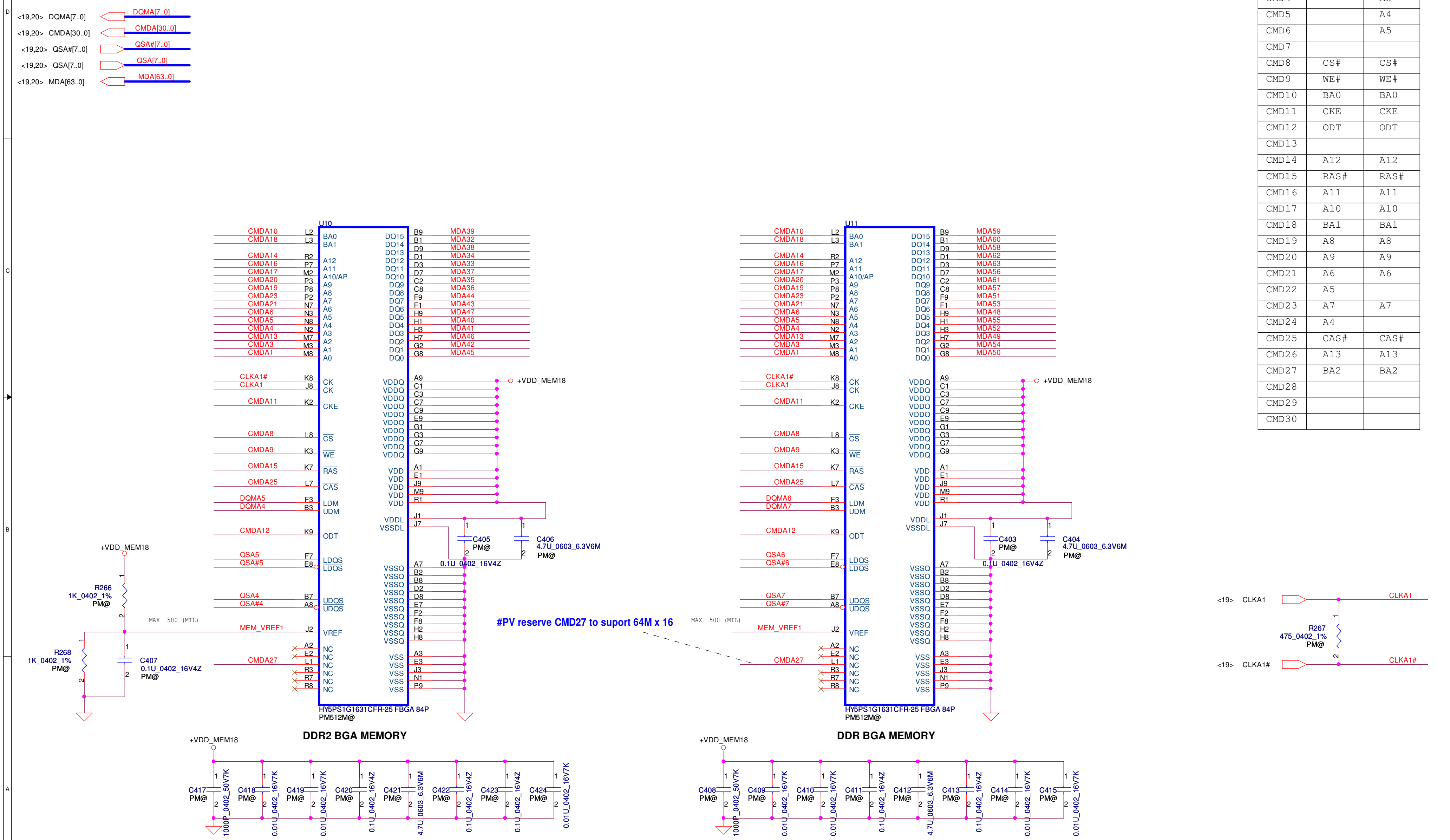
Date: Monday, April 27, 2009 | Sheet 7 of 16

VRAM DDR2 chips (256MB & 512MB)

32Mx16 DDR2 400MHz *4==>256MB

64Mx16 DDR2 400MHz*4==>512MB

	DATA Bus	
Address	0..31	32..63
CMD0	A3	
CMD1	A0	A0
CMD2	A2	
CMD3	A1	A1
CMD4		A3
CMD5		A4
CMD6		A5
CMD7		
CMD8	CS#	CS#
CMD9	WE#	WE#
CMD10	BA0	BA0
CMD11	CKE	CKE
CMD12	ODT	ODT
CMD13		
CMD14	A12	A12
CMD15	RAS#	RAS#
CMD16	A11	A11
CMD17	A10	A10
CMD18	BA1	BA1
CMD19	A8	A8
CMD20	A9	A9
CMD21	A6	A6
CMD22	A5	
CMD23	A7	A7
CMD24	A4	
CMD25	CAS#	CAS#
CMD26	A13	A13
CMD27	BA2	BA2
CMD28		
CMD29		
CMD30		

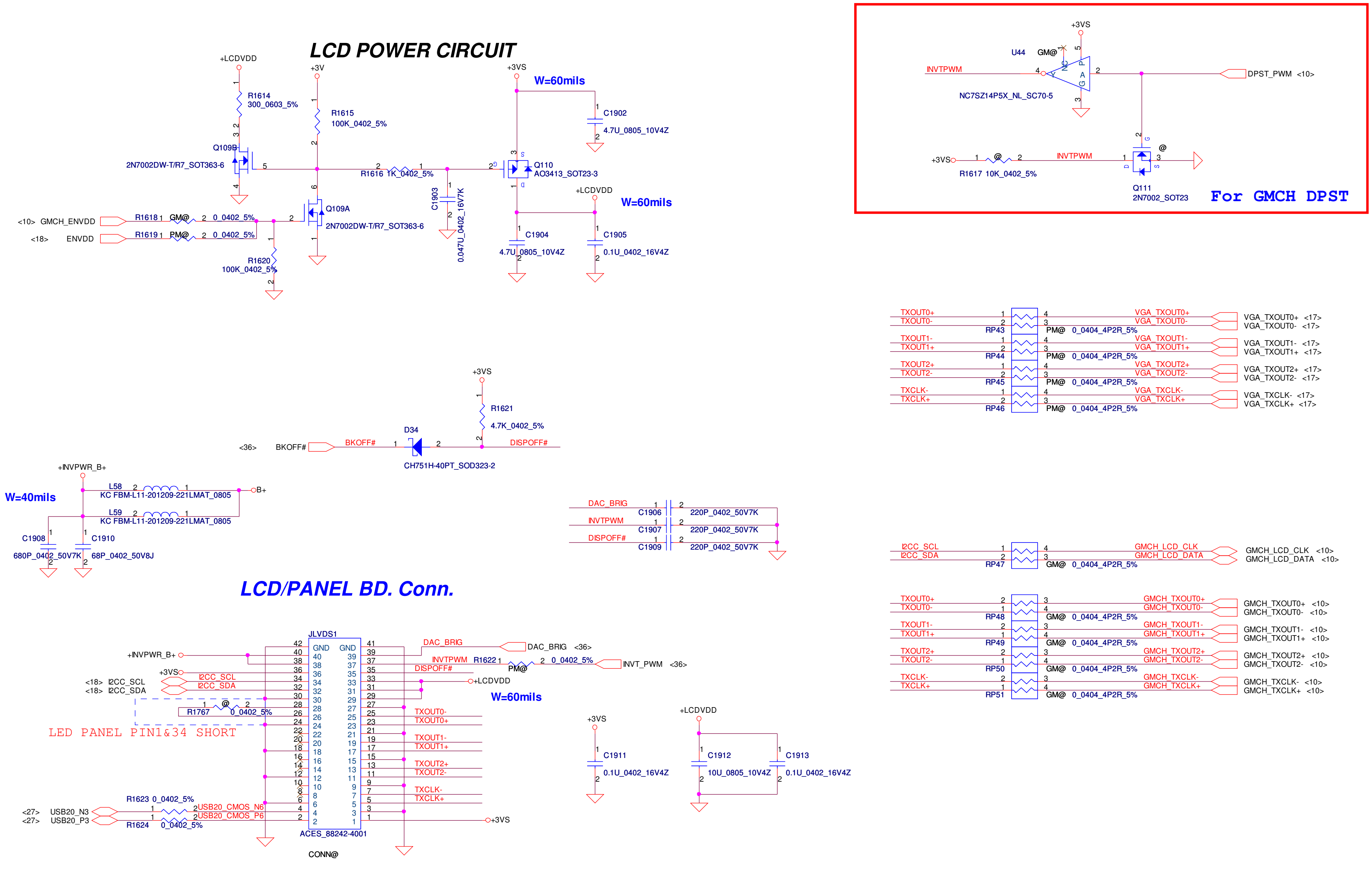


Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

Compal Electronics, Inc.

Title: VRAM 2

Size | Document Number: KAL90 | Rev 0.1



Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

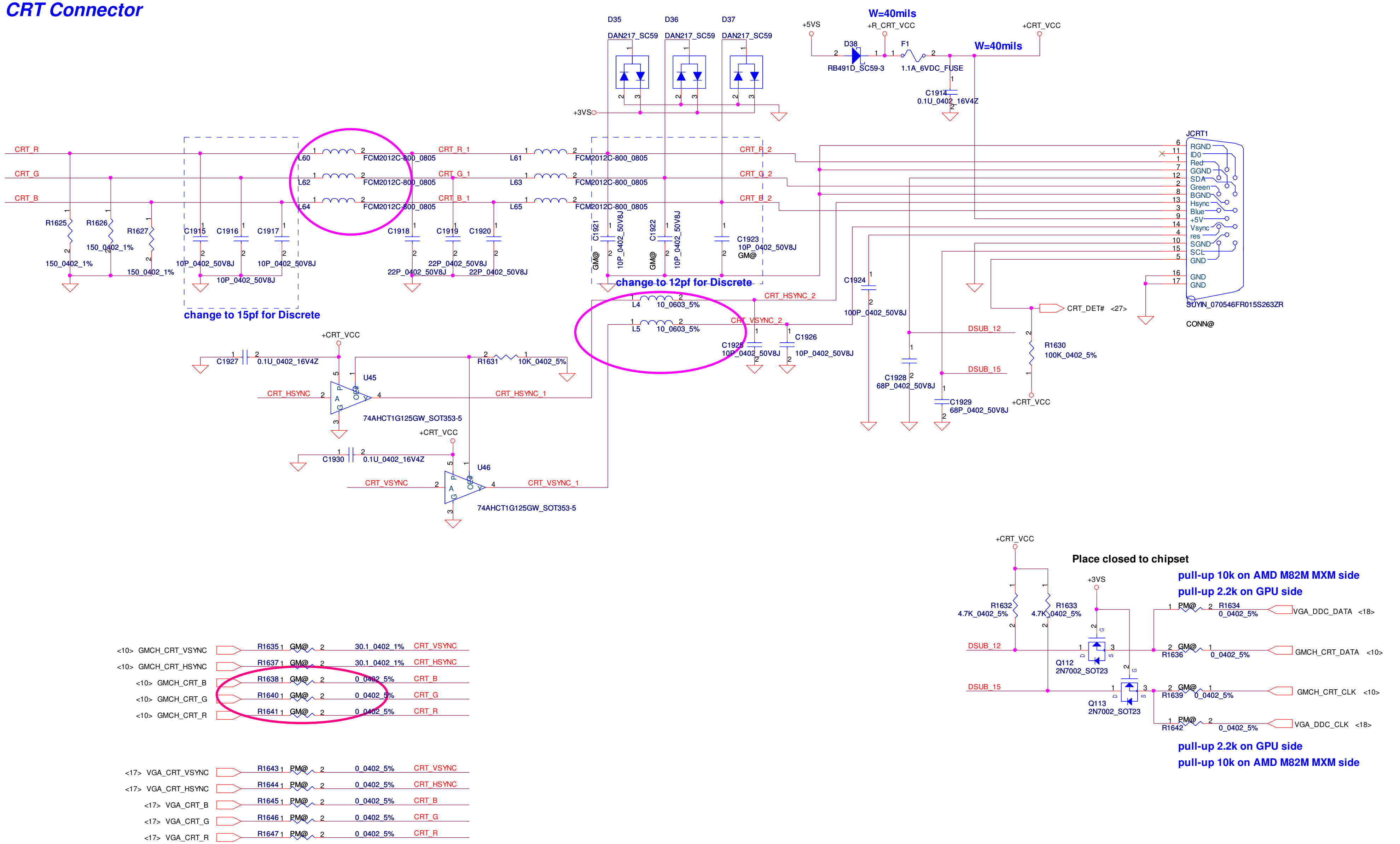
Compal Electronics, Inc.

Title: LVDS Connector

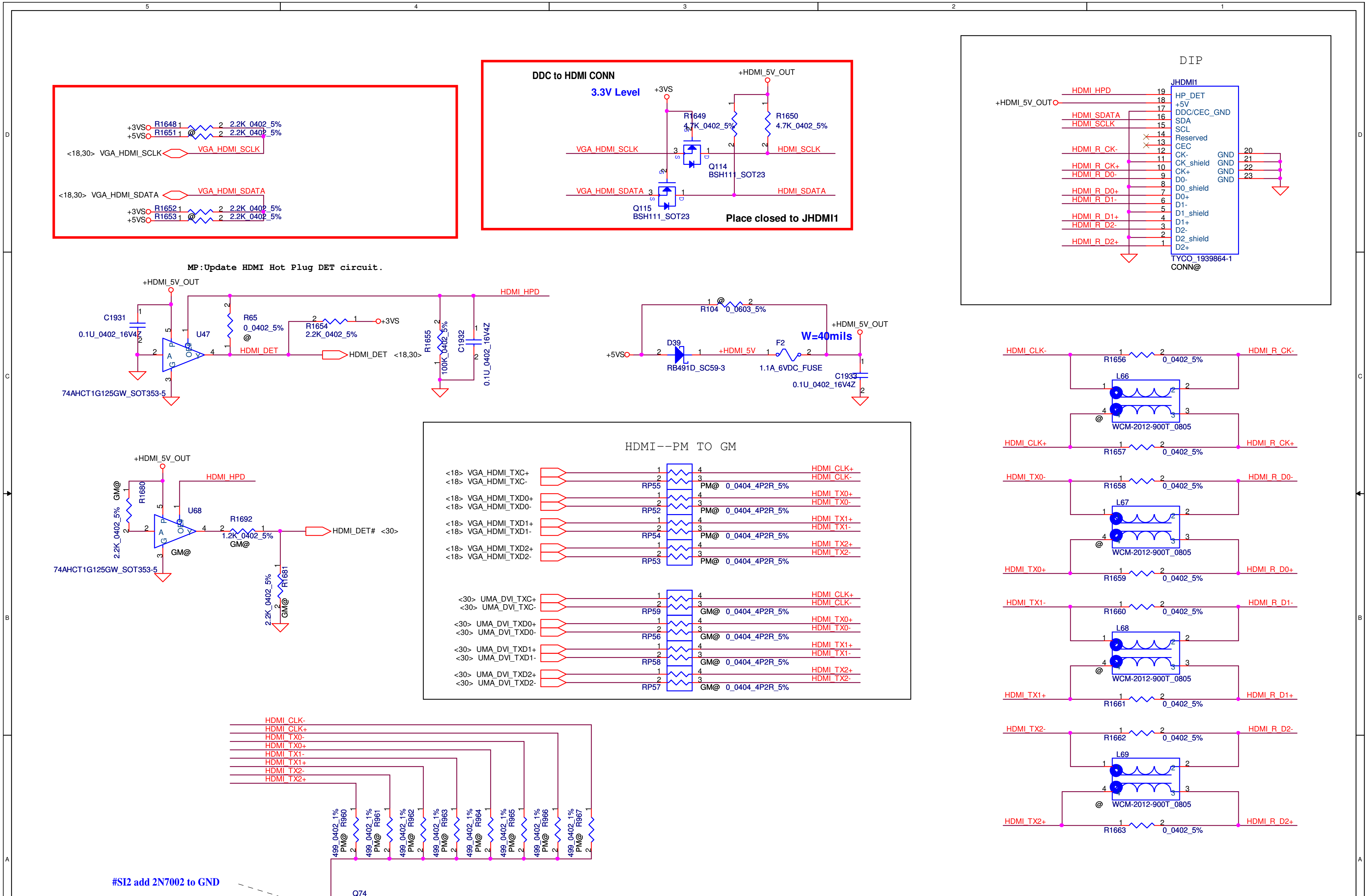
Size: Custom | Document Number: KALH0/KALG0/KAL90+ | Rev 1.0

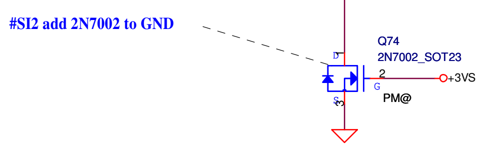
Date: Monday, April 27, 2009 | Sheet 22 of 53

CRT Connector



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: CRT Connector
				Size: B — Document Number: KALH0/KALG0/KAL90+ — Rev: 1.0
				Date: Monday, April 27, 2009 — Sheet 23 of 53

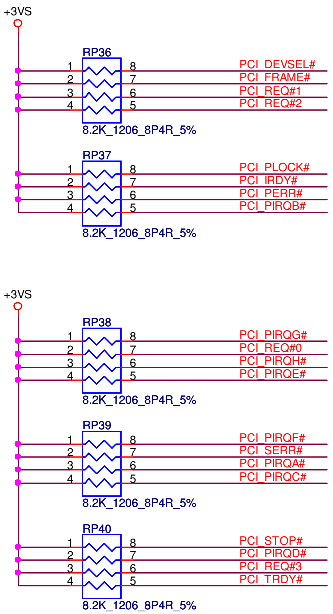


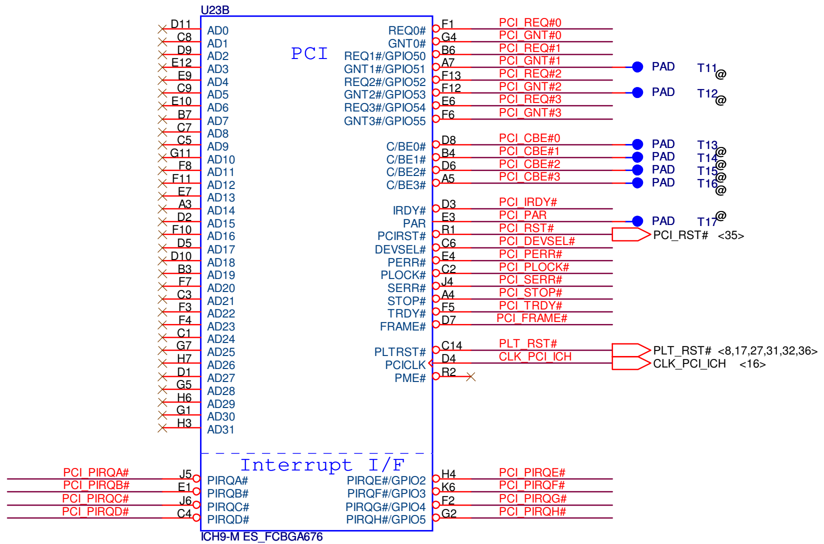


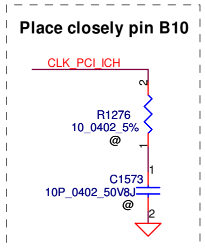
Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: HDMI Connector
				Size: Custom; Document Number: KALH0/KALG0/KAL90+; Rev: 1.0
				Date: Monday, April 27, 2009; Sheet 24 of 53

DMI for ESI-compatible operation

PCI_GNT#1	Low= DMI for ESI-compatible operation High= Default* (Internal pull-up)

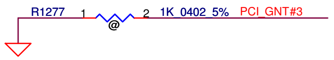


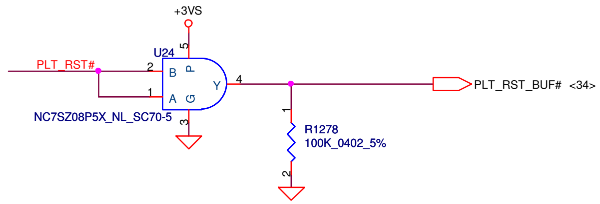




A16 Swap Override Strap

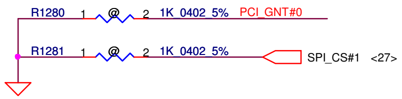
PCI_GNT#3	Low= A16 swap override Enable High= Default*





Boot BIOS Strap

PCI_GNT#0	SPI_CS#1	Boot BIOS Loaction
0	1	SPI
1	0	PCI
1	1	LPC*



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: ICH9M(1/4)-PCI
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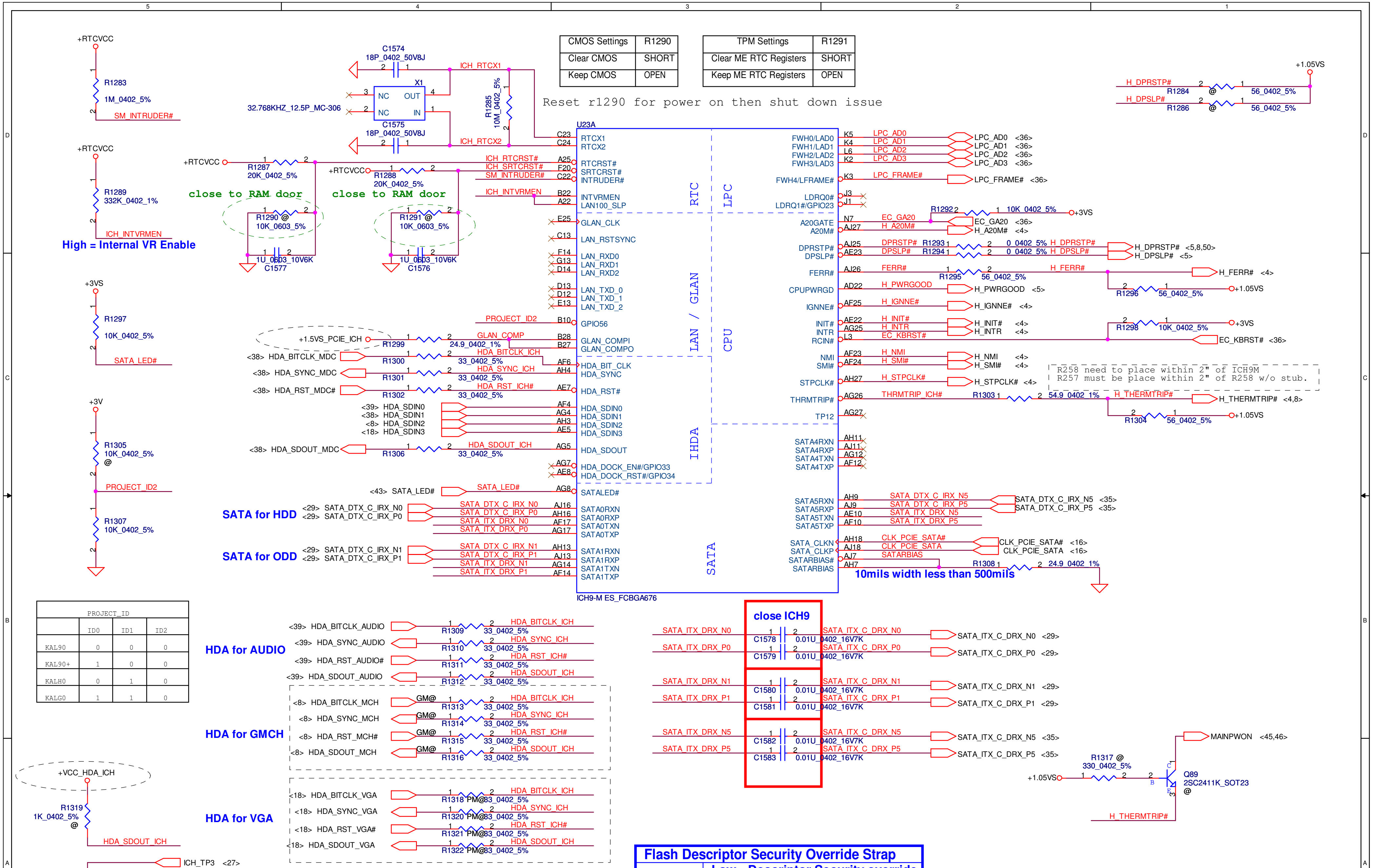
				Date: Monday, April 27, 2009 / Sheet 25 of 53



CMOS Settings	R1290
Clear CMOS	SHORT
Keep CMOS	OPEN

TPM Settings	R1291
Clear ME RTC Registers	SHORT
Keep ME RTC Registers	OPEN

PROJECT_ID			
	ID0	ID1	ID2
KAL90	0	0	0
KAL90+	1	0	0
KALH0	0	1	0
KALG0	1	1	0

XOR Chain Entrance Strap

ICH_TP3	HDA_SDOUT	Description
0	0	RSVD
0	1	Enter XOR Chain
1	0	Normal Operation
1	1	Set PCIE port config bit 1

Flash Descriptor Security Override Strap

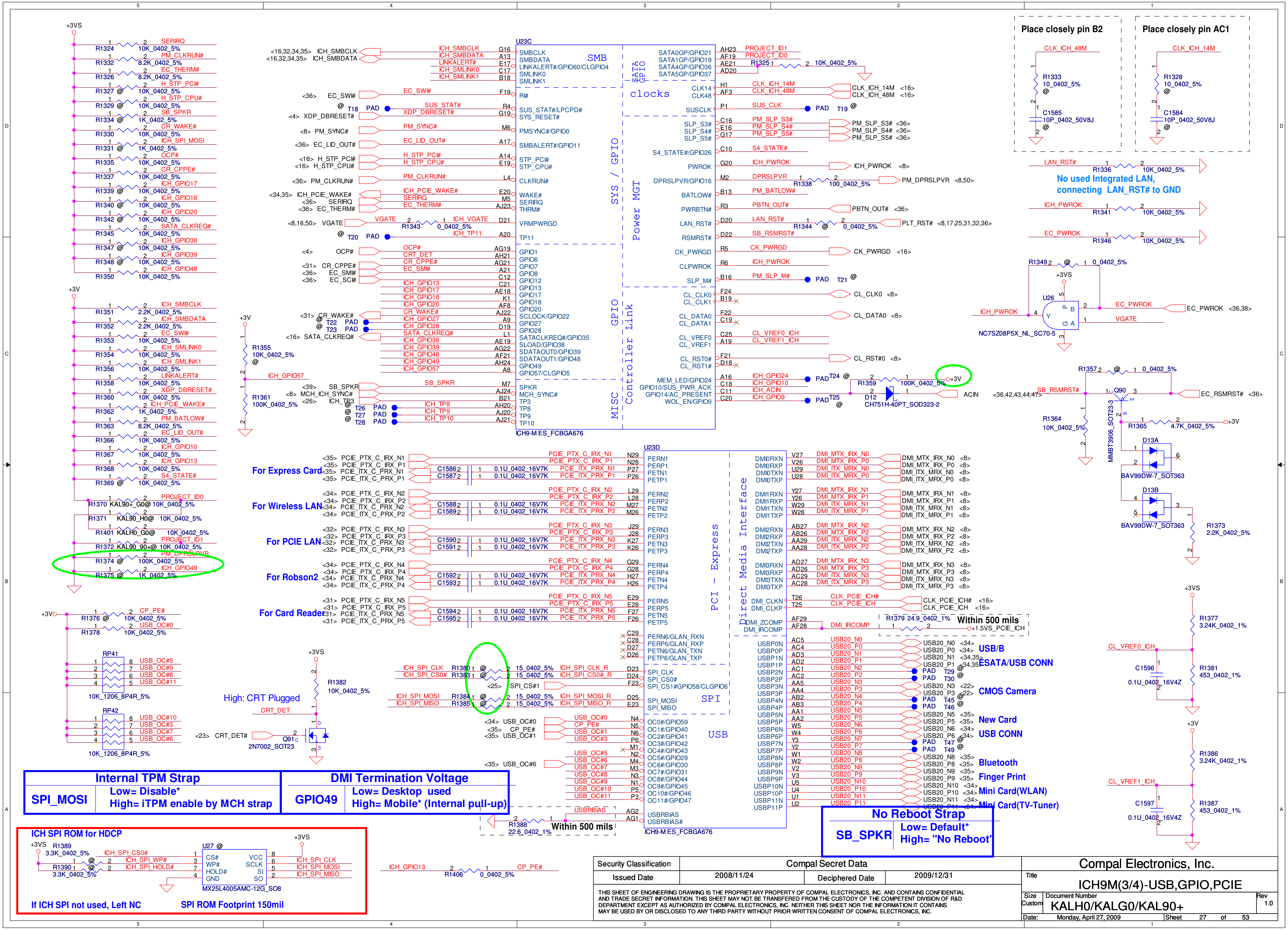
GPIO33	Low= Descriptor Security override High= Default* (Internal pull-up)

Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

Compal Electronics, Inc.

Title: ICH9M(2/4)-LAN,IDELPC,RTC

Size: Custom | Document Number: KALH0/KALG0/KAL90+ | Rev 1.0



Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

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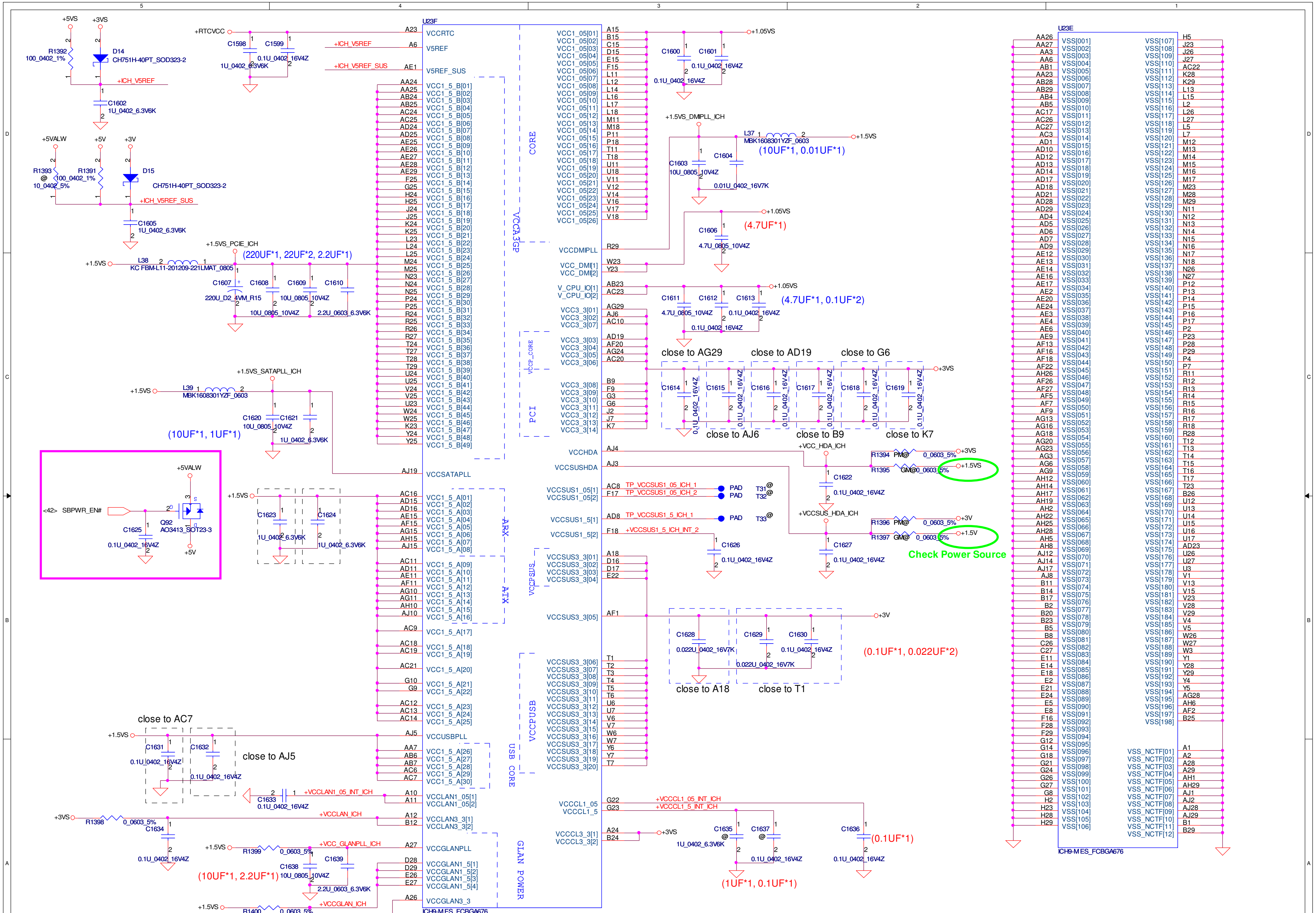
Compal Electronics, Inc.

Title: ICH9M(3/4)-USB,GPIO,PCIE

Size: Custom | Document Number: KALH0/KALG0/KAL90+ | Rev 1.0

Date: Monday, April 27, 2009 | Sheet 27 of 53



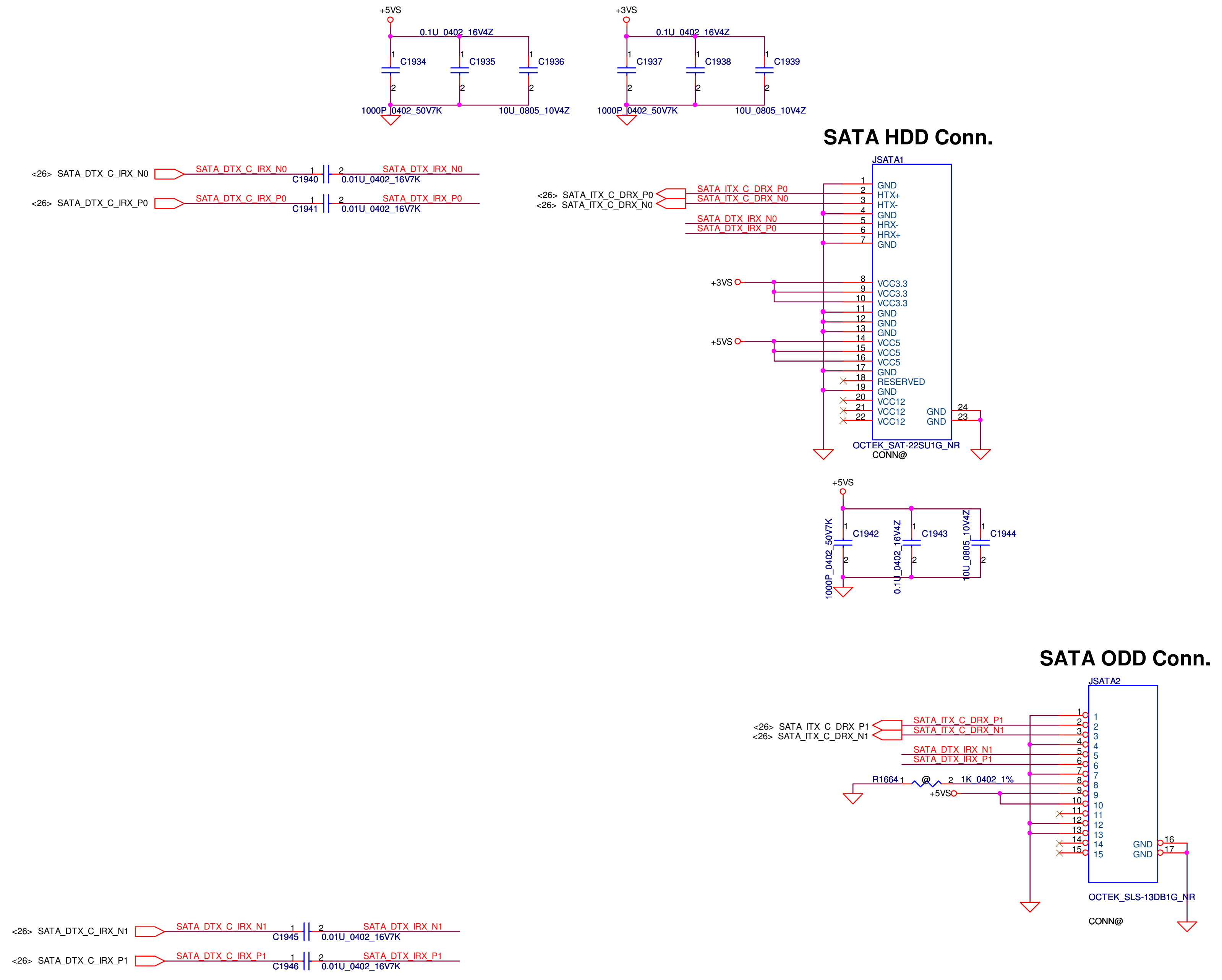
Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

Compal Electronics, Inc.

Title: ICH9M(4/4)-POWER&GND

Size: Custom | Document Number: KALH0/KALG0/KAL90+ | Rev 1.0

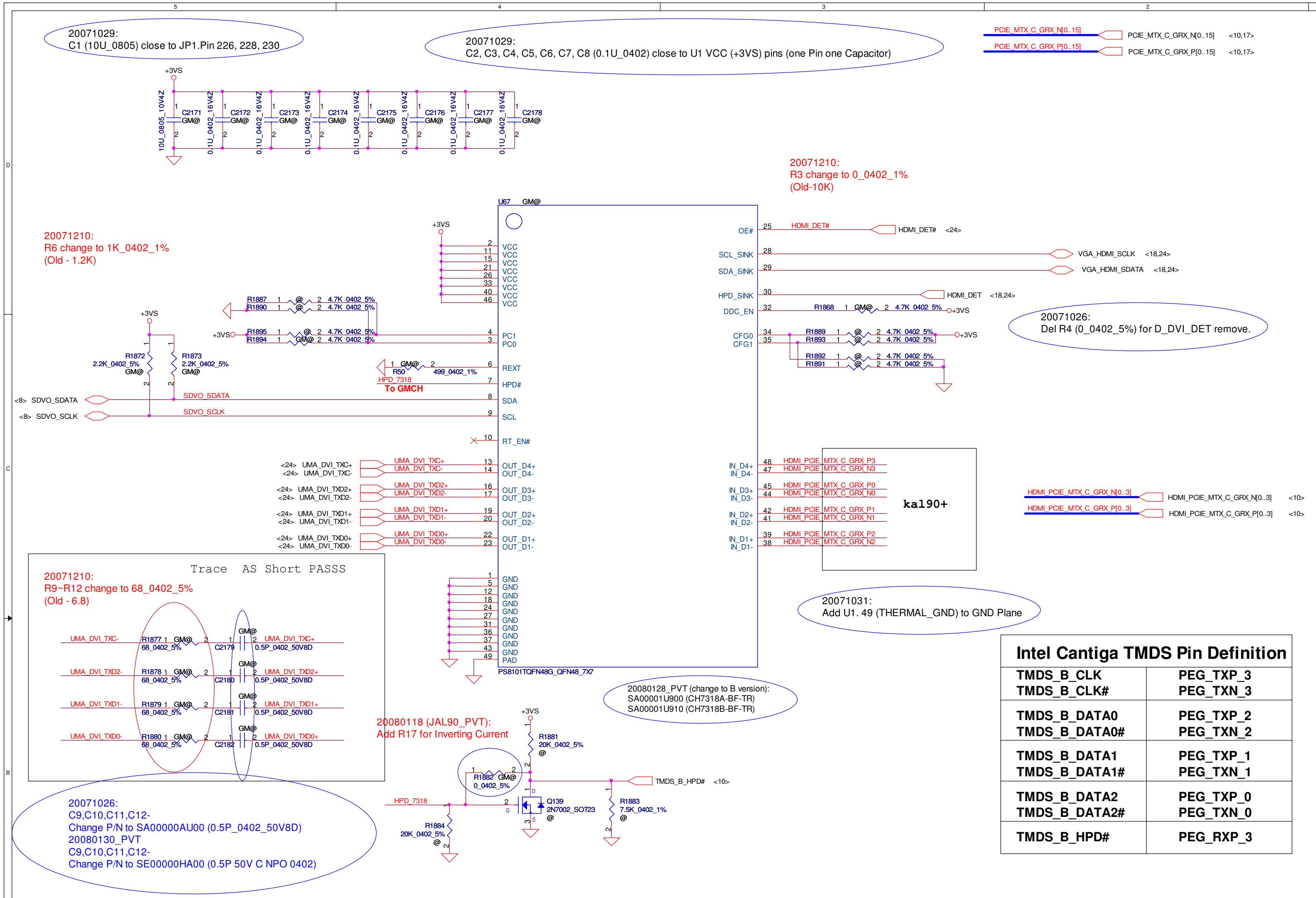
Date: Monday, April 27, 2009 | Sheet 28 of 53



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: HDD & ODD Connector

Size: Custom | Document Number: KALH0/KALG0/KAL90+ | Rev: 1.0

Date: Monday, April 27, 2009 |



Intel Cantiga TMDS Pin Definition

TMDS_B_CLK	PEG_TXP_3
TMDS_B_CLK#	PEG_TXN_3
TMDS_B_DATA0	PEG_TXP_2
TMDS_B_DATA0#	PEG_TXN_2
TMDS_B_DATA1	PEG_TXP_1
TMDS_B_DATA1#	PEG_TXN_1
TMDS_B_DATA2	PEG_TXP_0
TMDS_B_DATA2#	PEG_TXN_0
TMDS_B_HPD#	PEG_RXP_3

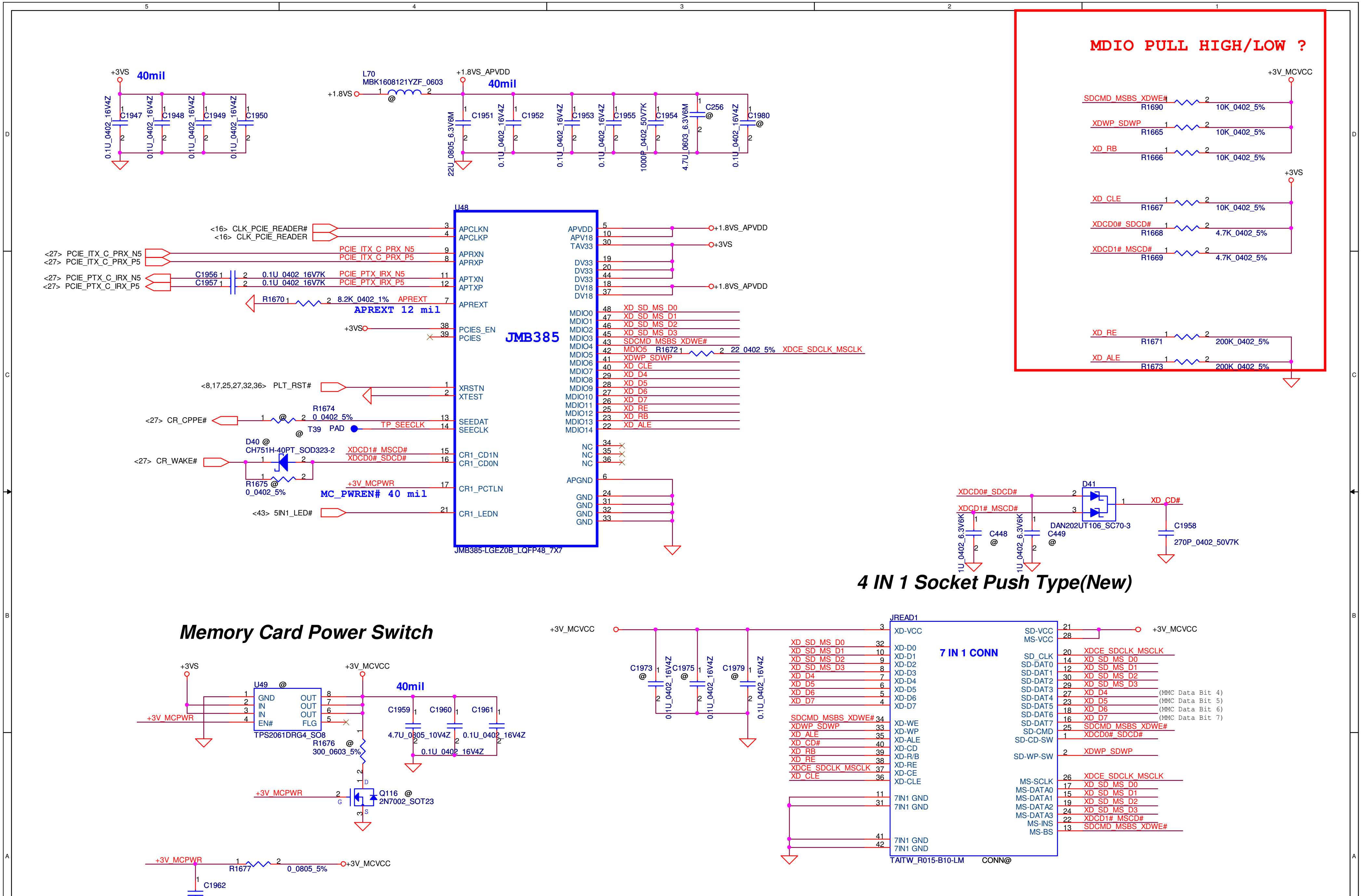
Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

Compal Electronics, Inc.

Title: HDMI/DVI CH7318

Size: C | Document Number: KALH0/KALG0/KAL90+ | Rev: 1.0

Date: Monday, April 27, 2009 |

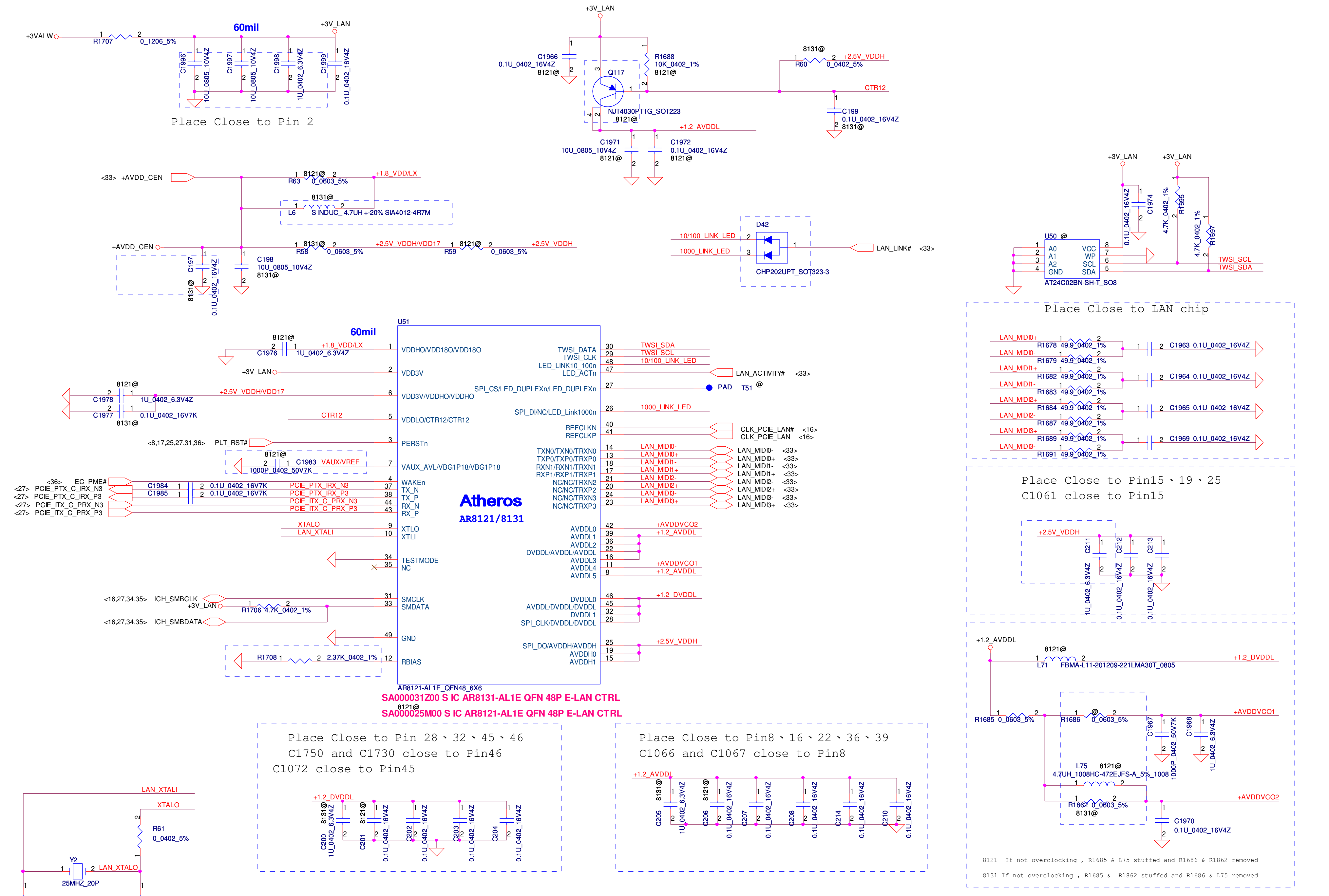


Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

Compal Electronics, Inc.

Title: Card Reader JMB385

Size	Document Number	Rev
Custom	KALH0/KALG0/KAL90+	1.0



Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

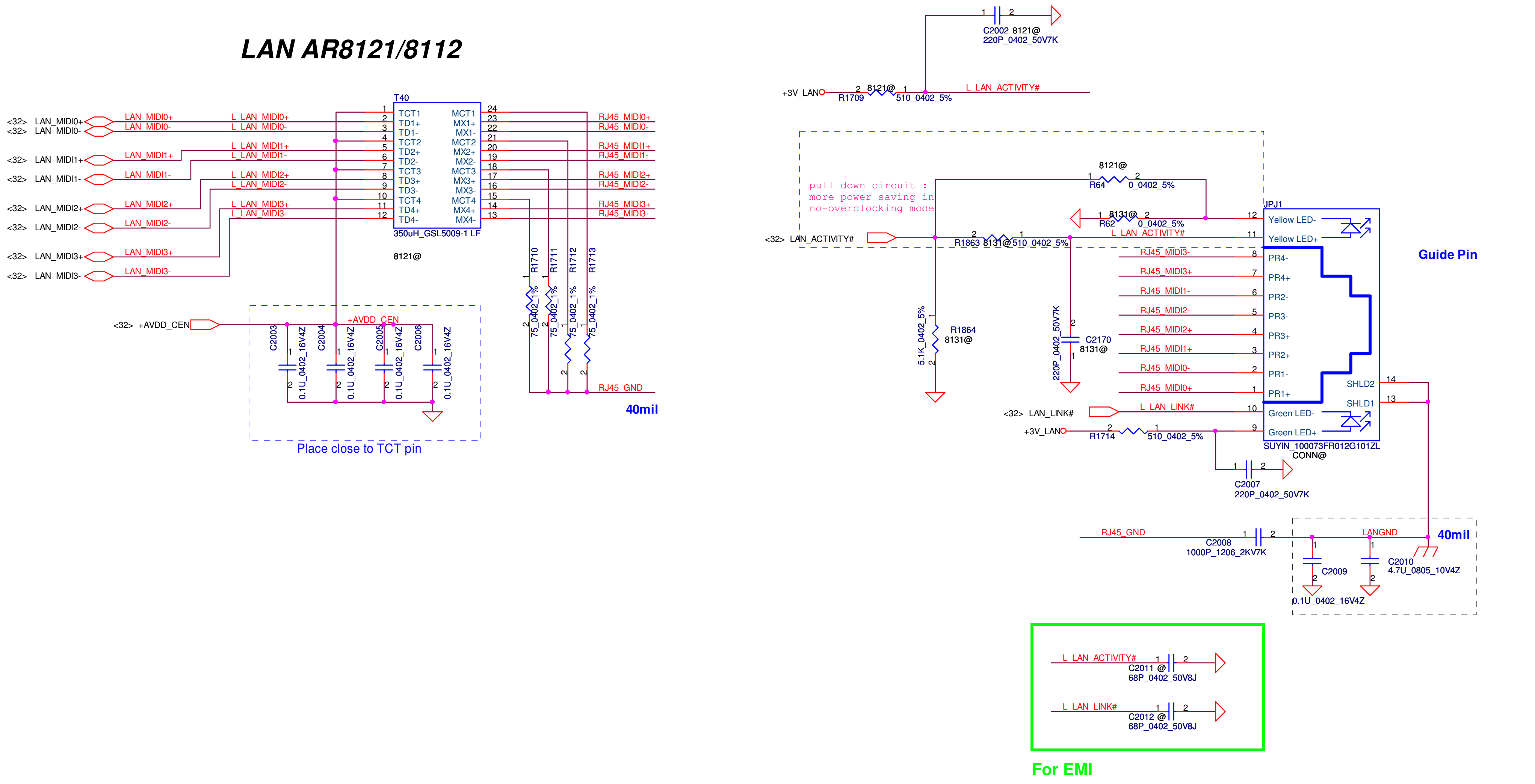
Compal Electronics, Inc.

Title: Atheros AR8131

Size: Custom | Document Number: KALH0/KALG0/KAL90+ | Rev: 1.0

Date: Monday, April 27, 2009 | Sheet 32 of 53

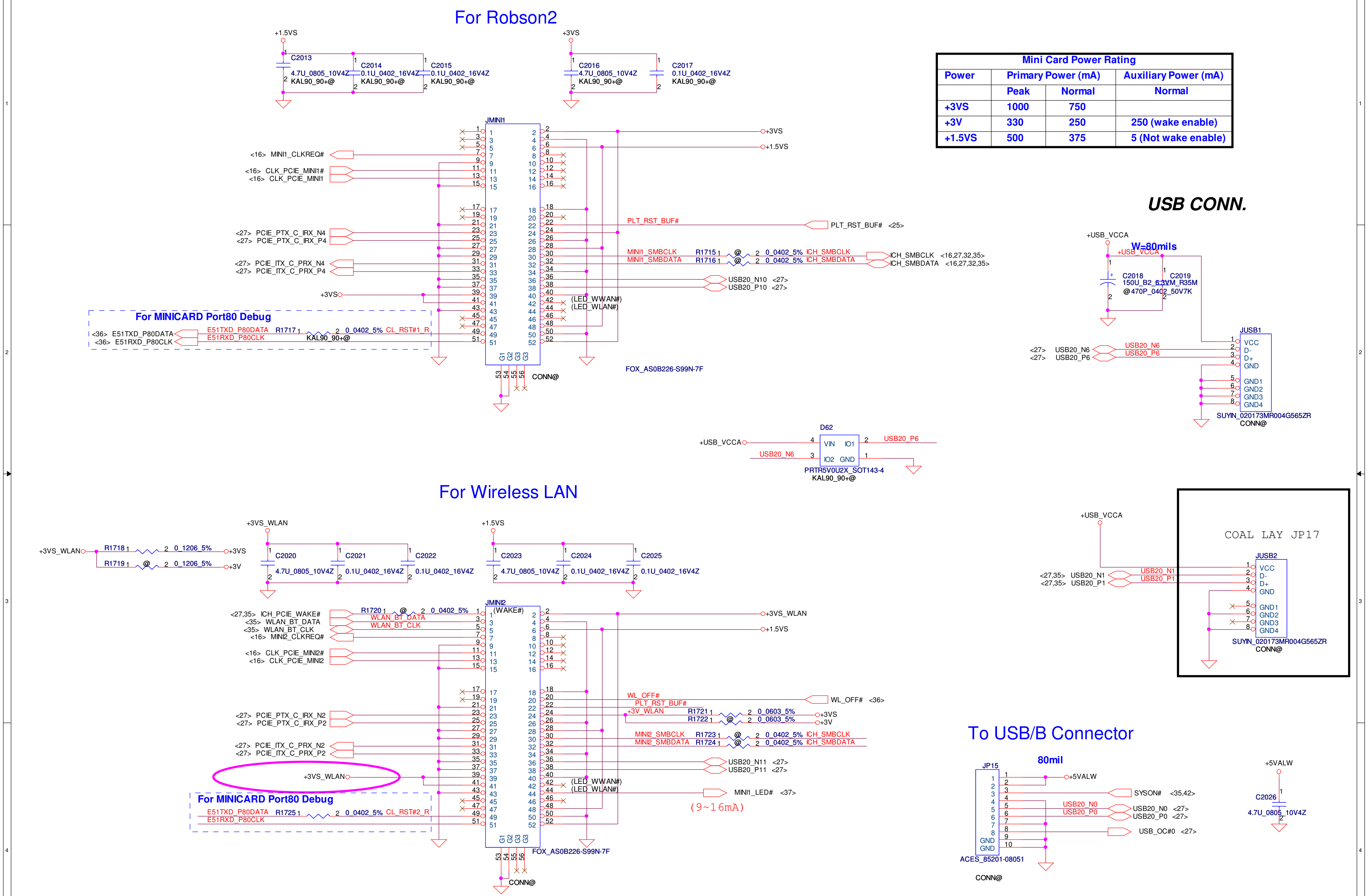
LAN AR8121/8112



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: LAN Magnetic & RJ45

Size	Document Number	Rev
B	KALH0/KALG0/KAL90+	1.0

Date: Monday, April 27, 2009 Sheet 33 of 53



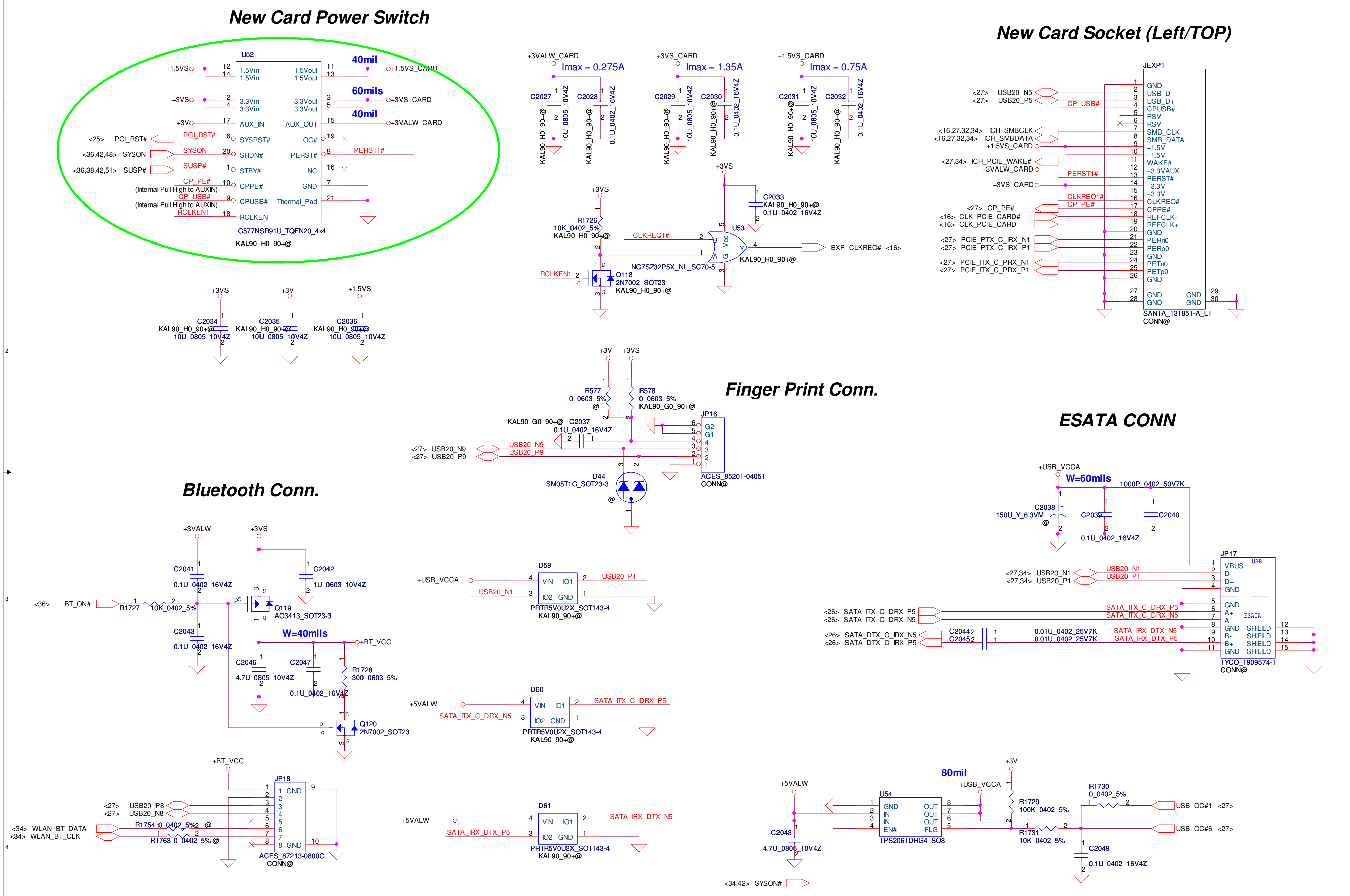
Mini Card Power Rating			
Power	Primary Power (mA)		Auxiliary Power (mA)
	Peak	Normal	Normal
+3VS	1000	750	
+3V	330	250	250 (wake enable)
+1.5VS	500	375	5 (Not wake enable)

Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

Compal Electronics, Inc.

Title: MINI CARD (WLAN & Robson2)

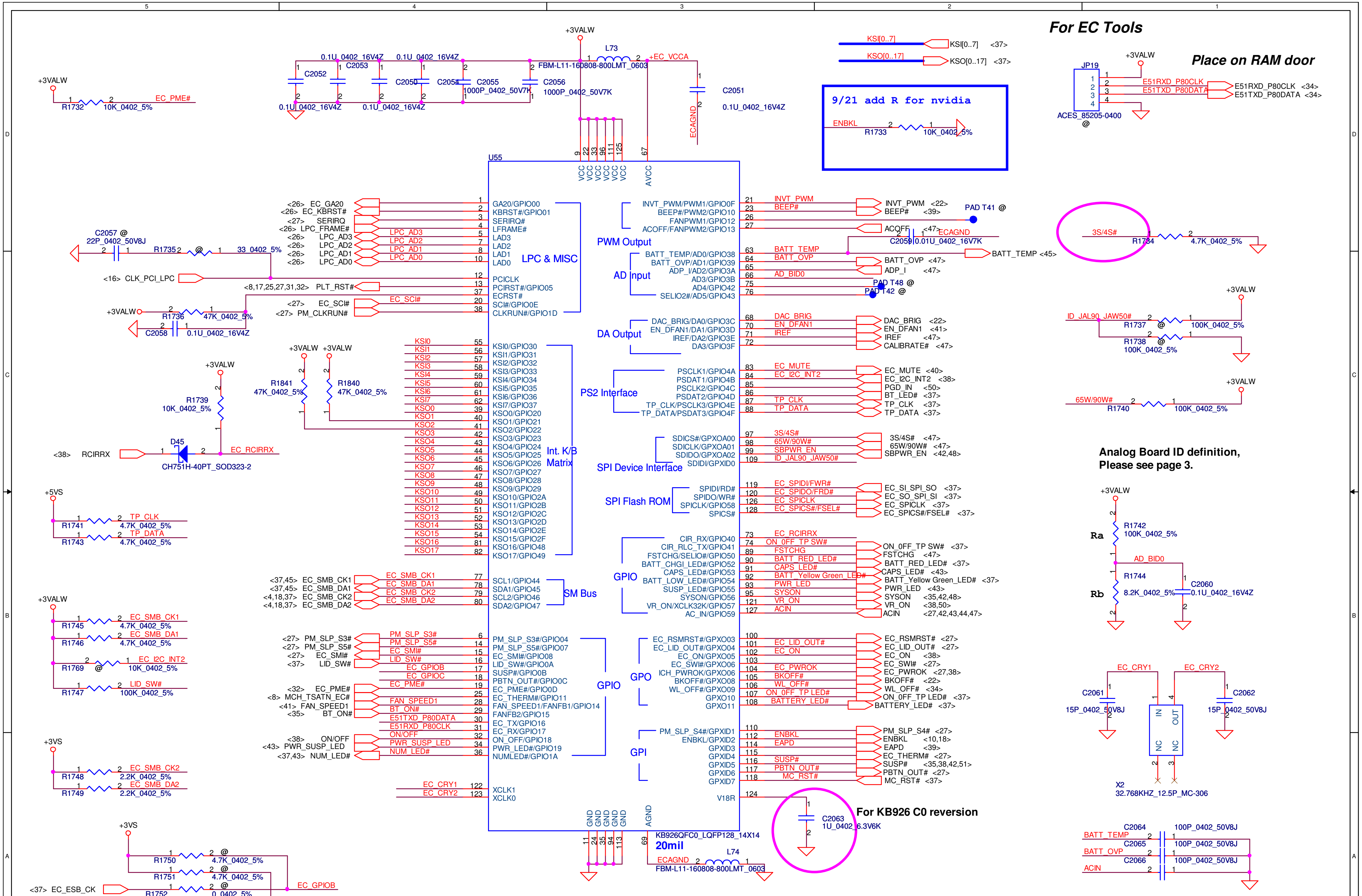
Size: B Document Number: KALH0/KALG0/KAL90+ Rev: 1.0



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: NEW CARD & eSATA Connector

Size	Document Number	Rev
B	KALH0/KALG0/KAL90+	1.0

Date: Monday, April 27, 2009



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: EC ENE KB926
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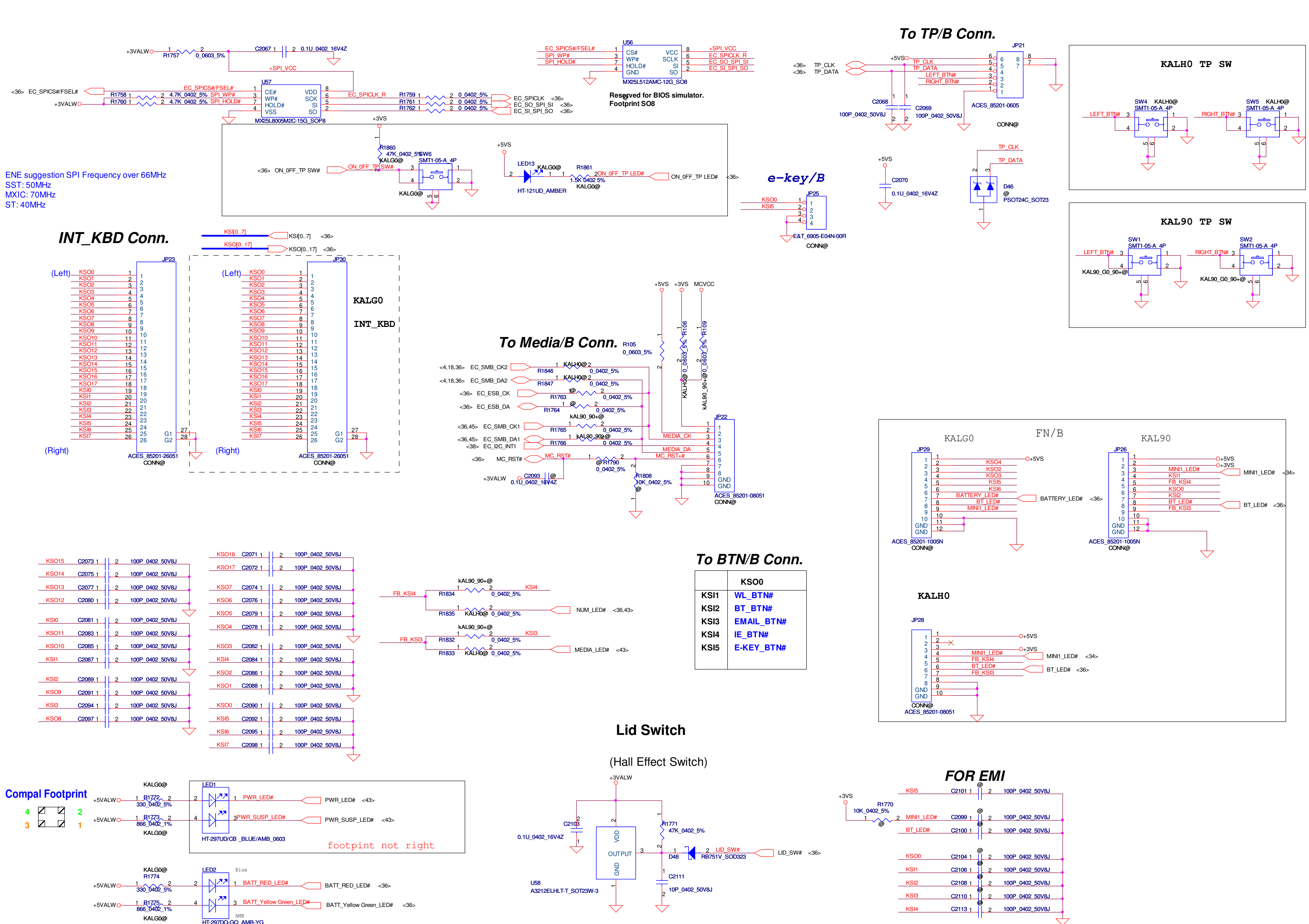
				Date: Monday, April 27, 2009; Sheet 36 of 53



Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

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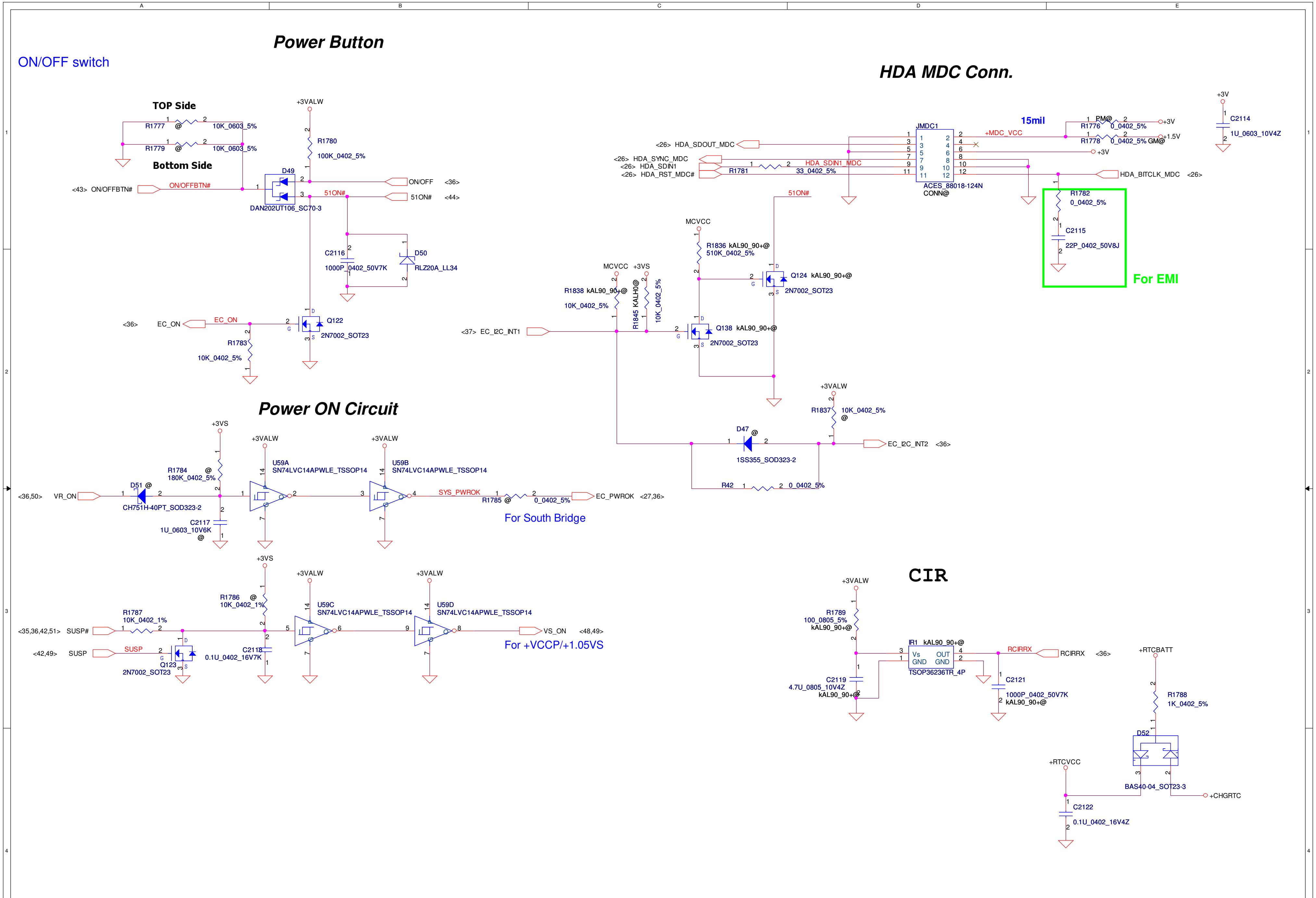
Compal Electronics, Inc.

Title: BIOS, I/O Port & K/B Connector

Size: C | Document Number: KALH0/KALG0/KAL90+ | Rev: 1.0

Date: Monday, April 27, 2009 | Sheet 37 of 53



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: Power OK, Reset,RTC, CIR, MDC
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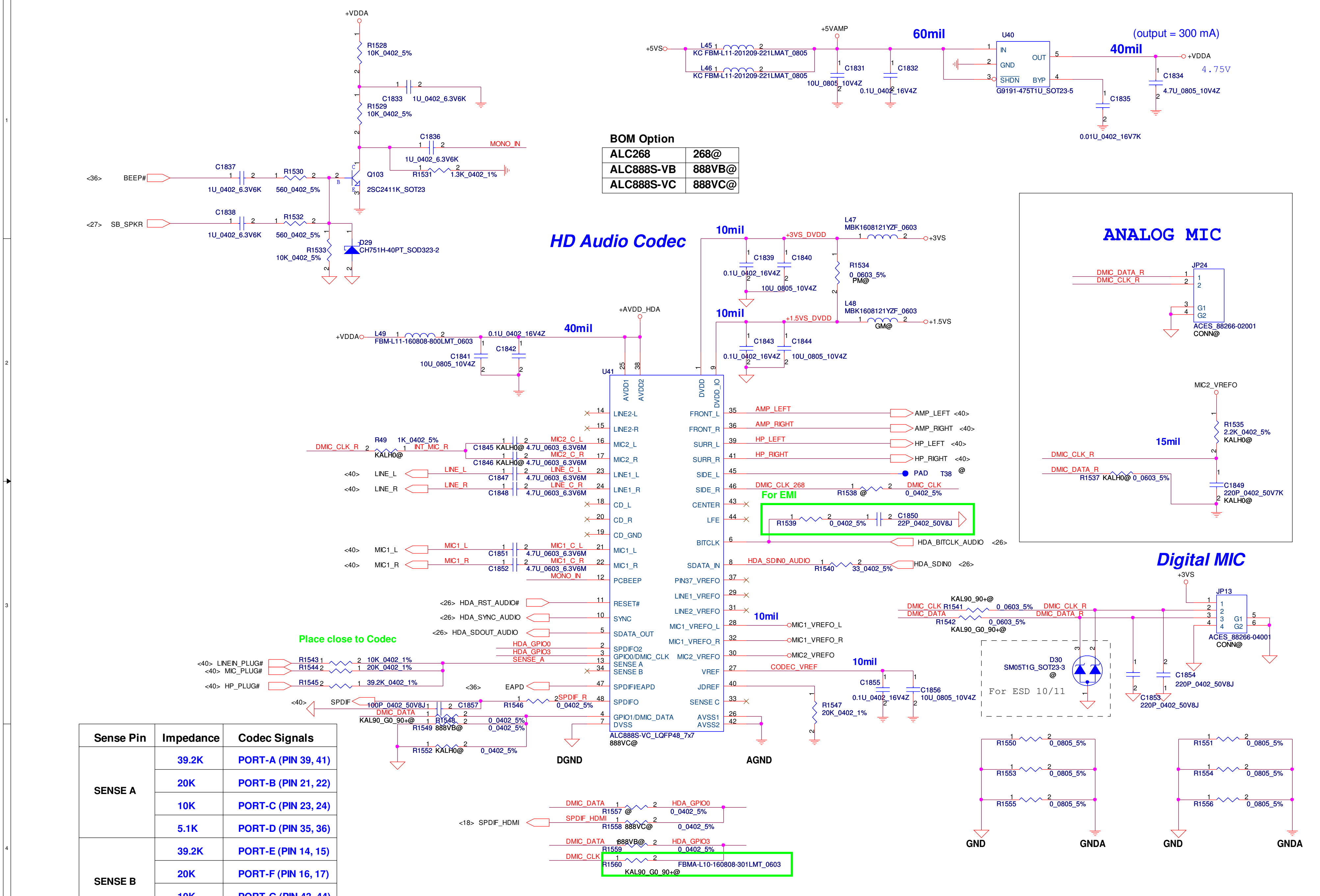
				Date: Monday, April 27, 2009; Sheet 38 of 53



BOM Option

ALC268	268@
ALC888S-VB	888VB@
ALC888S-VC	888VC@

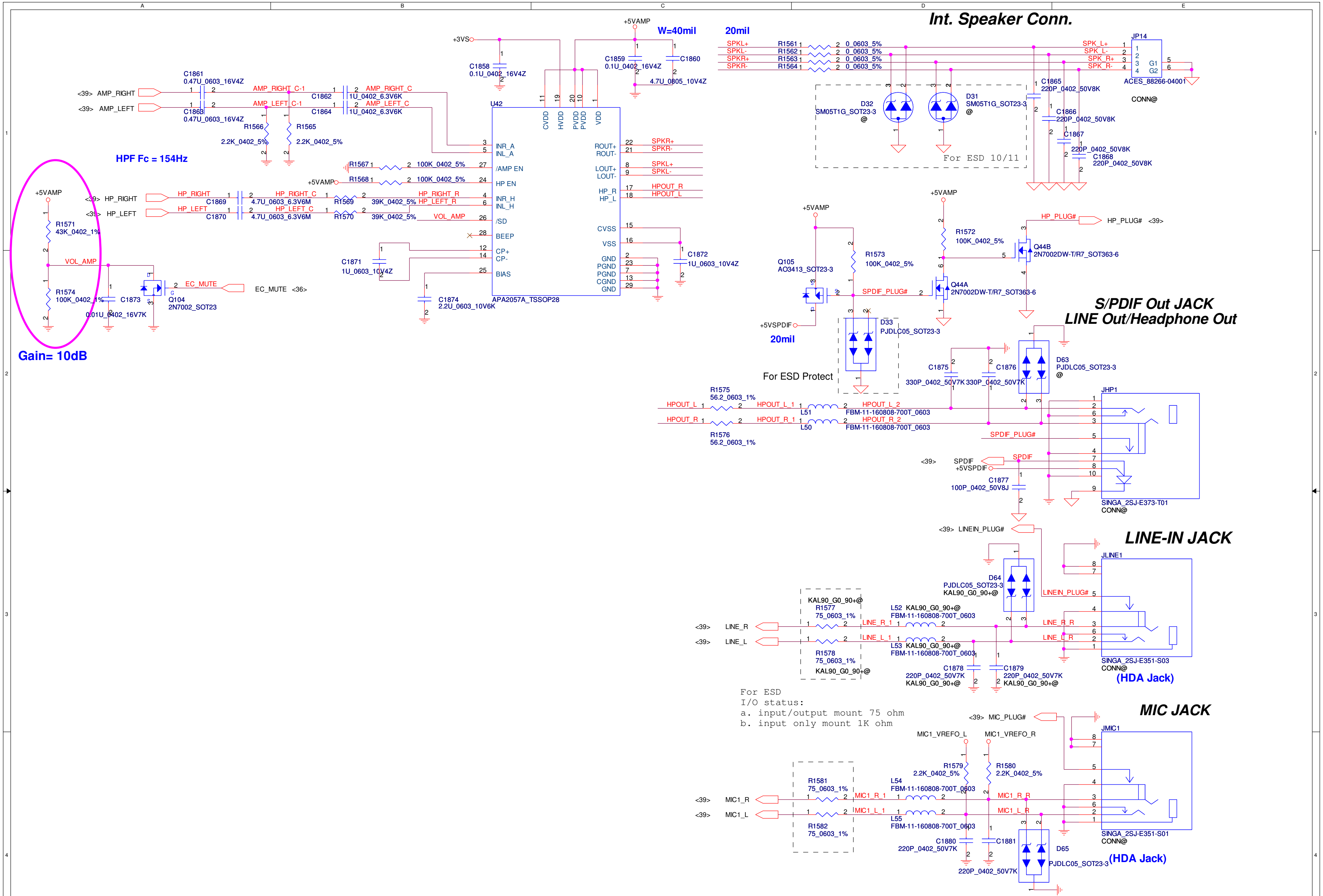
Sense Pin	Impedance	Codec Signals
SENSE A	39.2K	PORT-A (PIN 39, 41)
	20K	PORT-B (PIN 21, 22)
	10K	PORT-C (PIN 23, 24)
	5.1K	PORT-D (PIN 35, 36)
SENSE B	39.2K	PORT-E (PIN 14, 15)
	20K	PORT-F (PIN 16, 17)
	10K	PORT-G (PIN 43, 44)
	5.1K	PORT-H (PIN 45, 46)

Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

Compal Electronics, Inc.

Title: HD Audio Codec ALC888S-VC

Size: B | Document Number: KALH0/KALG0/KAL90+ | Rev: 1.0



Security Classification	Compal Secret Data		
Issued Date	2008/11/24	Deciphered Date	2009/12/31

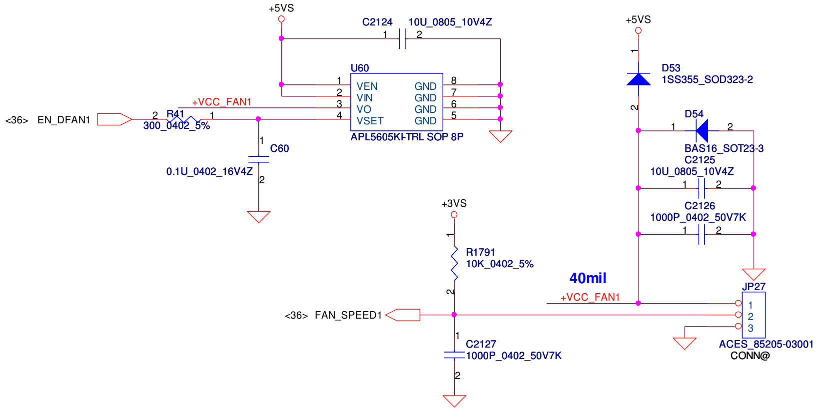
Compal Electronics, Inc.

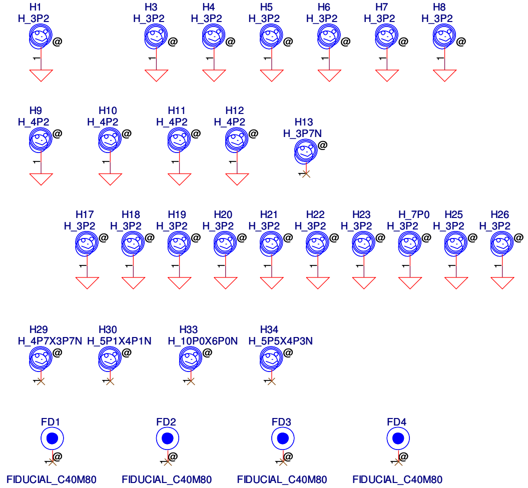
Title: Amplifier & Audio Jack

Size	Document Number	Rev
B	KALH0/KALG0/KAL90+	1.0

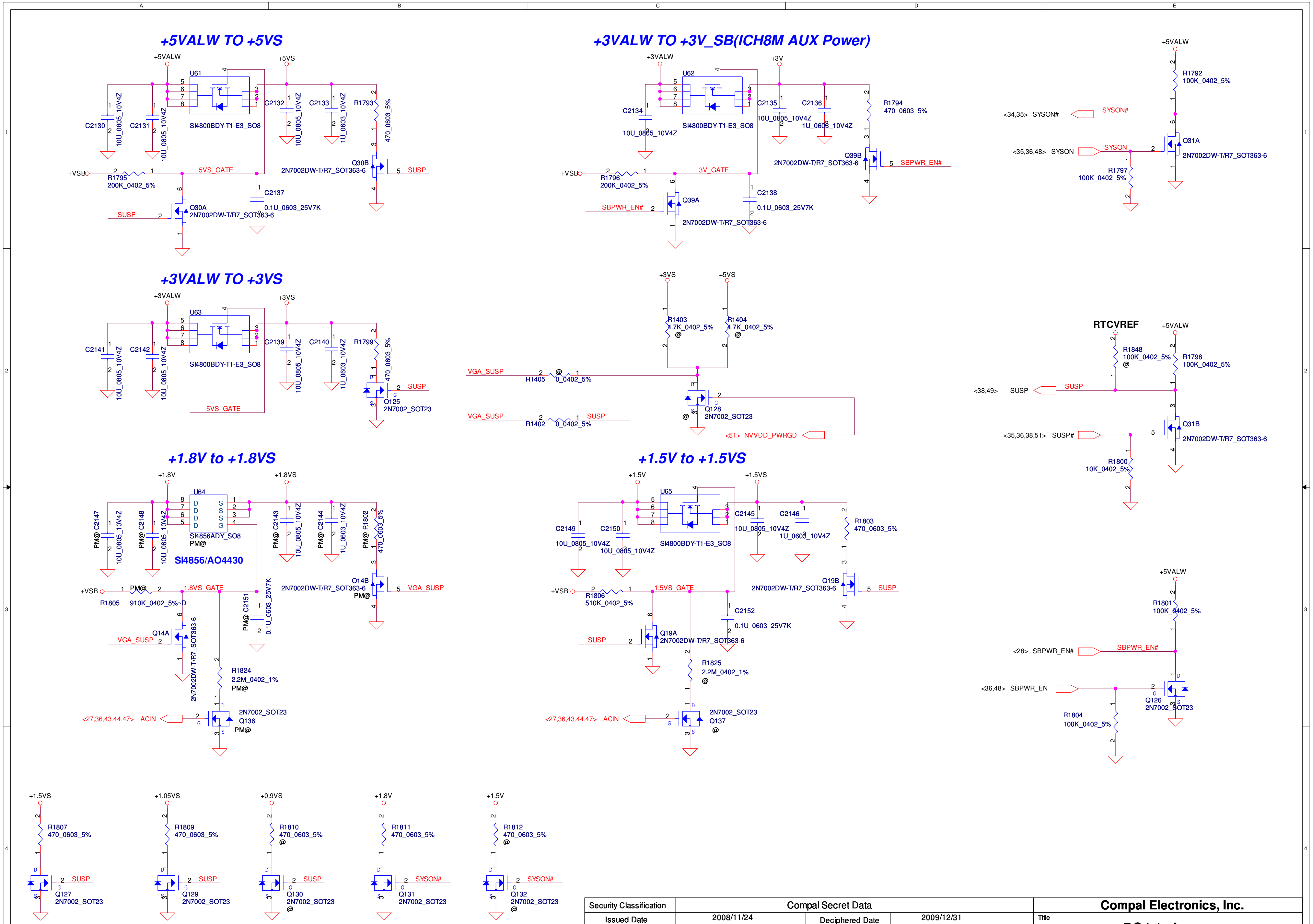
Date: Monday, April 27, 2009

FAN1 Conn





Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: FAN & COVER LIGHT& Screw Hole
				Size B; Document Number KALH0/KALG0/KAL90+; Rev 1.0
				Date: Monday, April 27, 2009; Sheet 41 of 53



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: DC Interface
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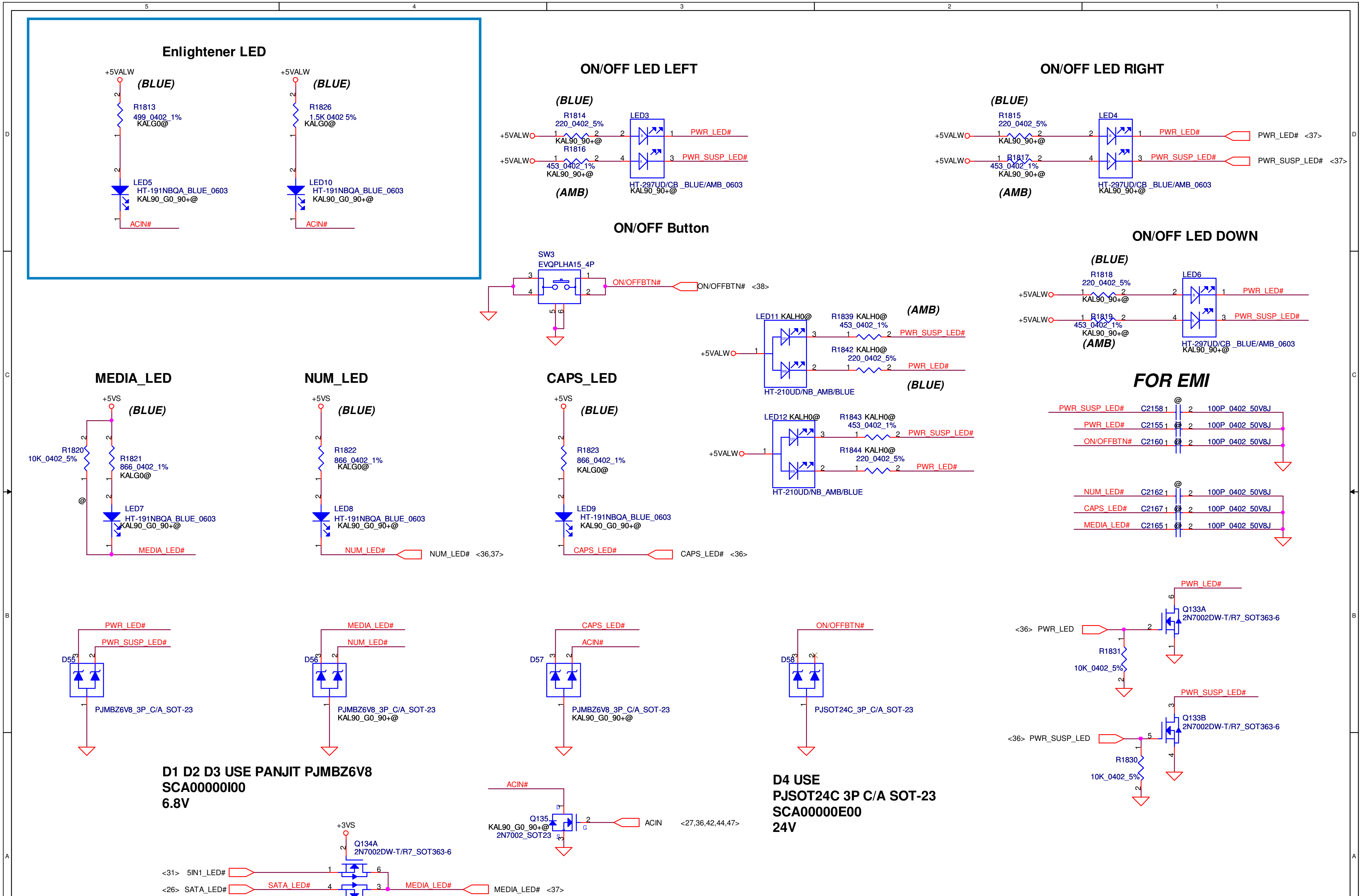
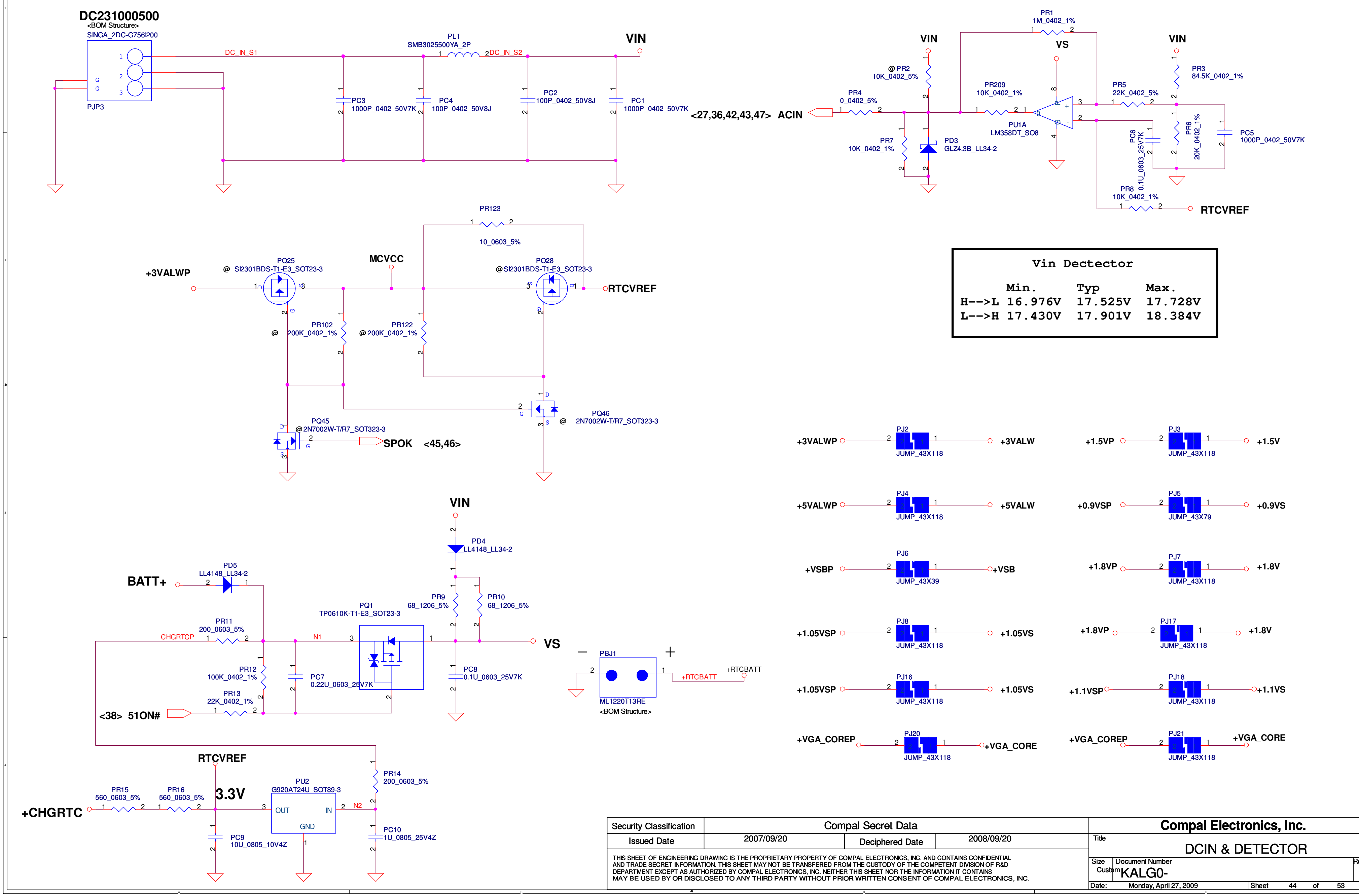
				Date: Monday, April 27, 2009; Sheet 42 of 53



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title: PWR/B
				Size: Custom; Document Number: KALH0/KALG0/KAL90+; Rev 1.0
				Date: Monday, April 27, 2009; Sheet 43 of 53



Vin Dectector

	Min.	Typ	Max.
H-->L	16.976V	17.525V	17.728V
L-->H	17.430V	17.901V	18.384V

Security Classification	Compal Secret Data		
Issued Date	2007/09/20	Deciphered Date	2008/09/20

Compal Electronics, Inc.

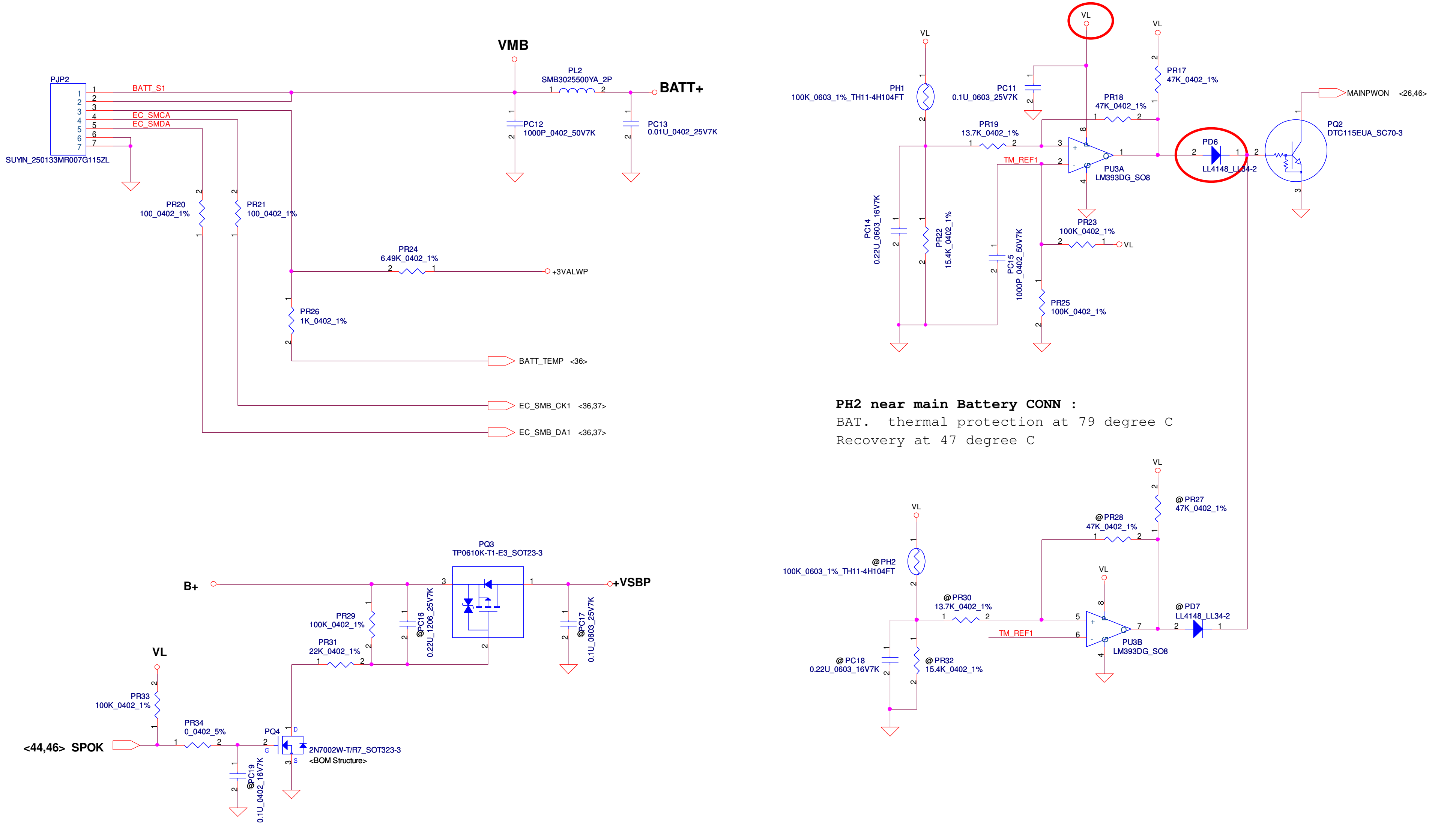
Title

DCIN & DETECTOR

Size Custom | Document Number KALG0- | Rev 0.1

PH1 under CPU botten side :
CPU thermal protection at 96 degree C
Recovery at 60 degree C

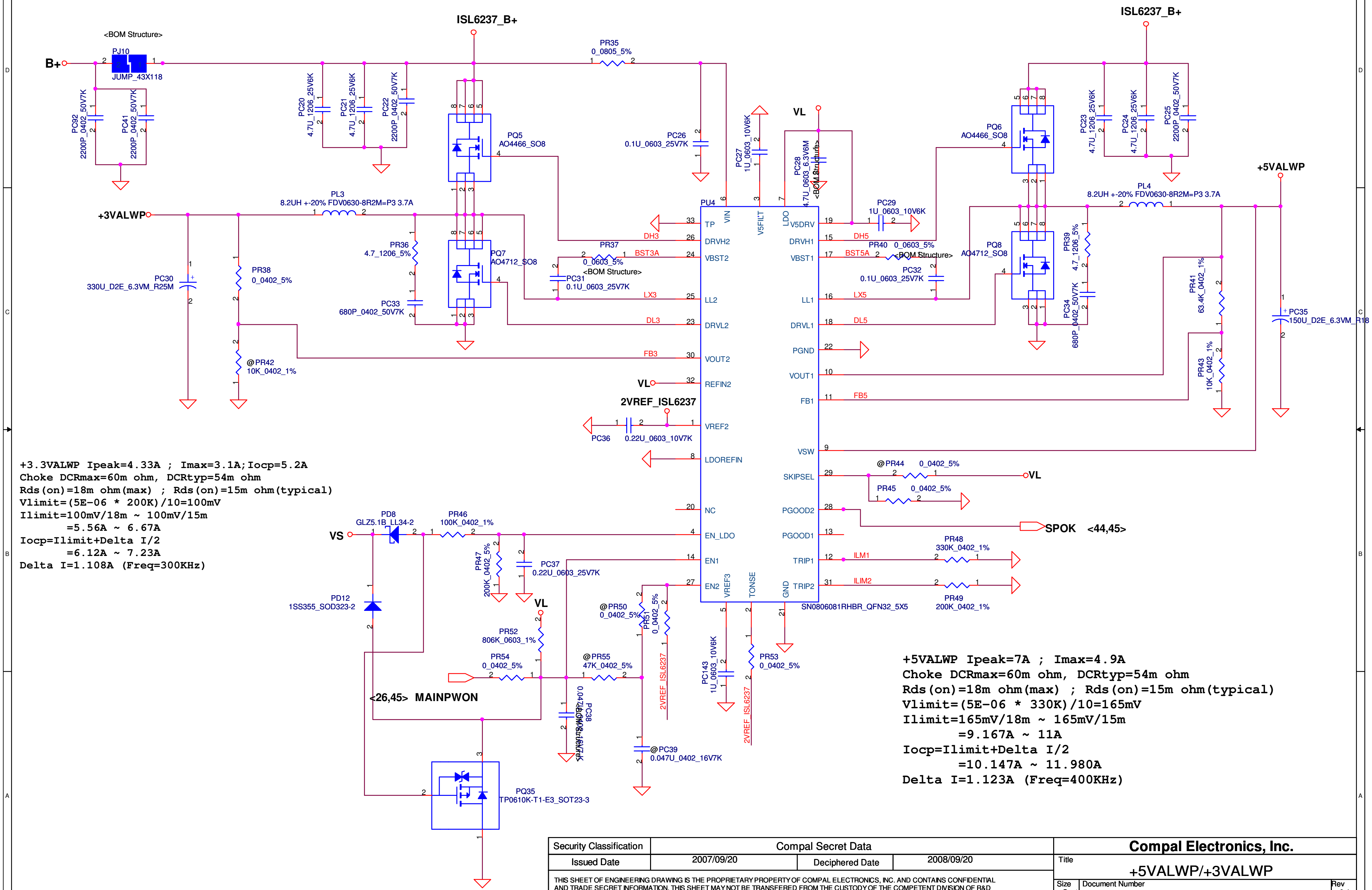
PH2 near main Battery CONN :
BAT. thermal protection at 79 degree C
Recovery at 47 degree C



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2007/09/20	Deciphered Date	2008/09/20	Title: BATTERY CONN / OTP

Size: Custom | Document Number: KALG0- | Rev: 0.1

Date: Monday, April 27, 2009 |



+3.3VALWP Ipeak=4.33A ; Imax=3.1A;Iocp=5.2A
Choke DCRmax=60m ohm, DCRtyp=54m ohm
Rds(on)=18m ohm(max) ; Rds(on)=15m ohm(typical)
Vlimit=(5E-06 * 200K)/10=100mV
Ilimit=100mV/18m ~ 100mV/15m
=5.56A ~ 6.67A
Iocp=Ilimit+Delta I/2
=6.12A ~ 7.23A
Delta I=1.108A (Freq=300KHz)

+5VALWP Ipeak=7A ; Imax=4.9A
Choke DCRmax=60m ohm, DCRtyp=54m ohm
Rds(on)=18m ohm(max) ; Rds(on)=15m ohm(typical)
Vlimit=(5E-06 * 330K)/10=165mV
Ilimit=165mV/18m ~ 165mV/15m
=9.167A ~ 11A
Iocp=Ilimit+Delta I/2
=10.147A ~ 11.980A
Delta I=1.123A (Freq=400KHz)

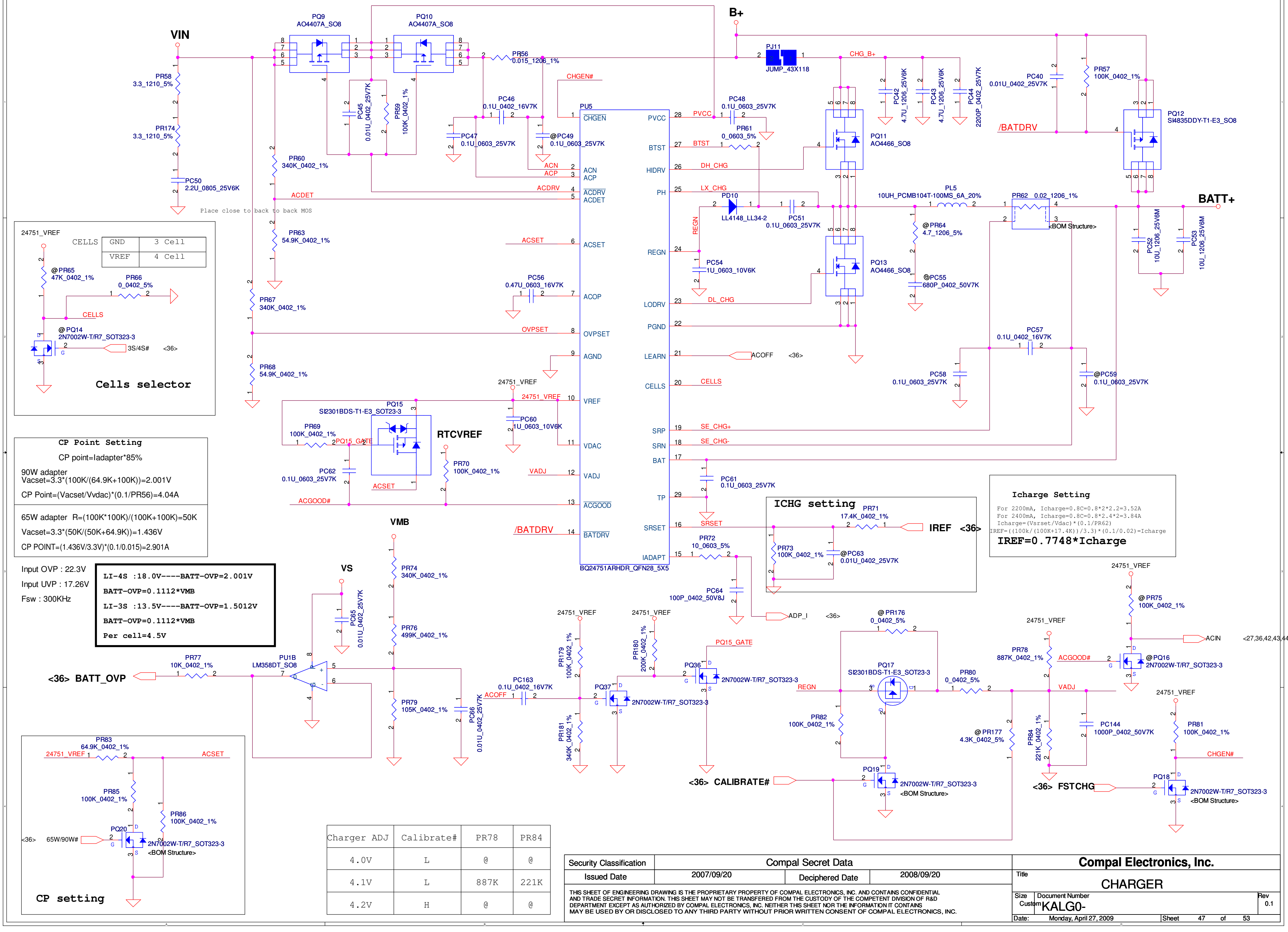
Security Classification	Compal Secret Data		
Issued Date	2007/09/20	Deciphered Date	2008/09/20

Compal Electronics, Inc.

Title: +5VALWP/+3VALWP

Size: Custom | Document Number: KALG0- | Rev: 0.1

Date: Monday, April 27, 2009 |



CELLS	
GND	3 Cell
VREF	4 Cell

CP Point Setting

CP point=Iadapter*85%

90W adapter
Vacset=3.3*(100K/(64.9K+100K))=2.001V

CP Point=(Vacset/Vvdac)*(0.1/PR56)=4.04A

65W adapter R=(100K*100K)/(100K+100K)=50K

Vacset=3.3*(50K/(50K+64.9K))=1.436V

CP POINT=(1.436V/3.3V)*(0.1/0.015)=2.901A

Input OVP : 22.3V
Input UVP : 17.26V
Fsw : 300KHz

LI-4S :18.0V----BATT-OVP=2.001V
BATT-OVP=0.1112*VMB
LI-3S :13.5V----BATT-OVP=1.5012V
BATT-OVP=0.1112*VMB
Per cell=4.5V

Icharge Setting

For 2200mA, Icharge=0.8C=0.8*2.2=3.52A
For 2400mA, Icharge=0.8C=0.8*2.4*2=3.84A
Icharge=(Vsrset/Vdac)*(0.1/PR62)
IREF=((100k/(100K+17.4K))/3.3)*(0.1/0.02)=Icharge
IREF=0.7748*Icharge

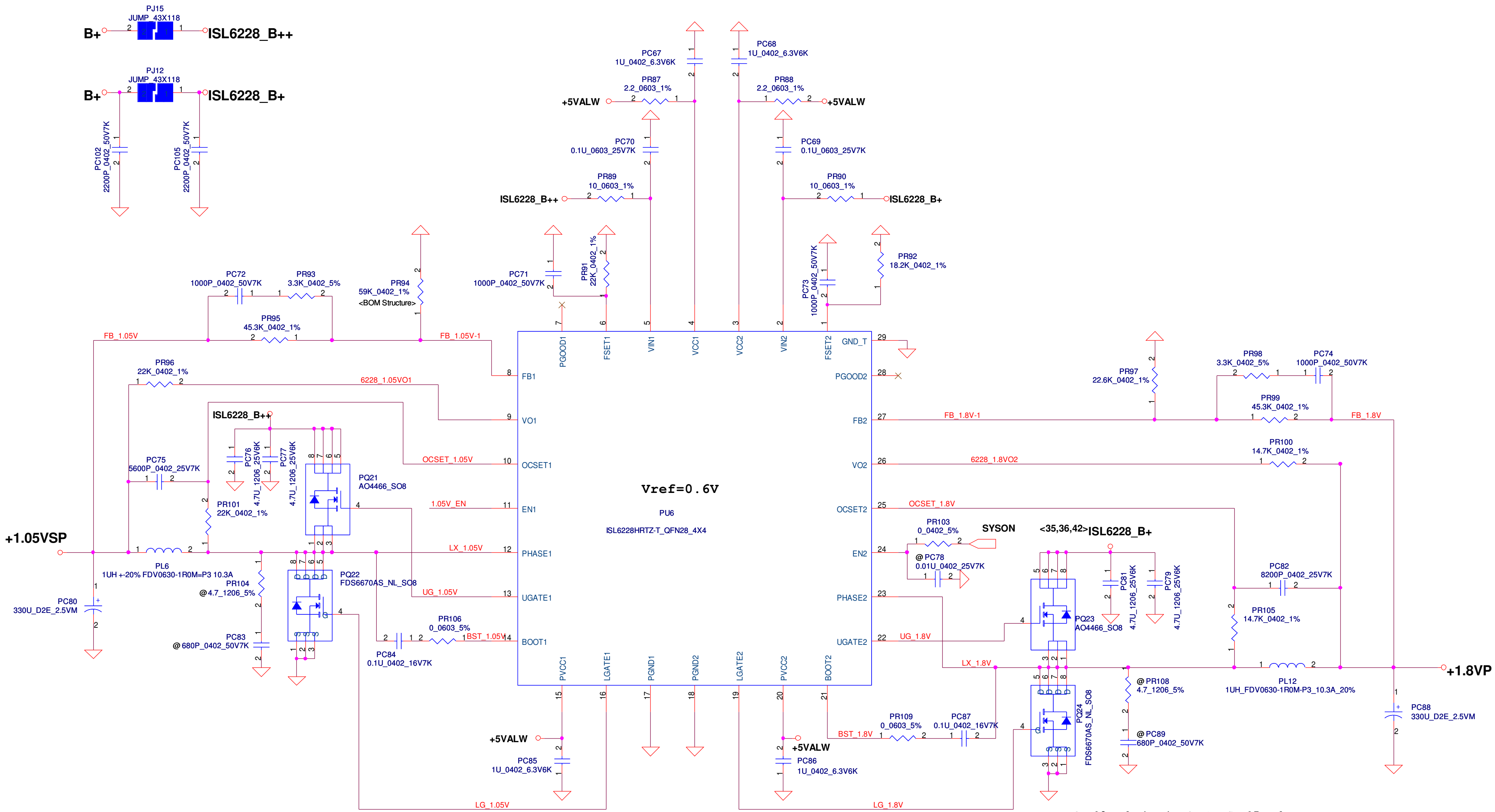
Charger ADJ	Calibrate#	PR78	PR84
4.0V	L	@	@
4.1V	L	887K	221K
4.2V	H	@	@

Security Classification	Compal Secret Data		
Issued Date	2007/09/20	Deciphered Date	2008/09/20

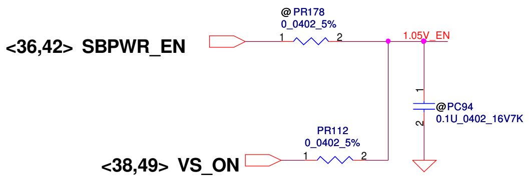
Compal Electronics, Inc.

Title: CHARGER

Size: Custom | Document Number: KALG0- | Rev: 0.1




```
DCR 10m ohm(max)   Cout ESR=15m ohm
1.05VSP ( Ibudget=17.47A)   OCP Seting
Fsw=1/1.5E-10*22k =303K
Vo=Vref*((PR95+PR94)/PR94)
Ipeak=13.29A, Imax=9.31A
Iocp=13.29*1.2=15.95A
Delta I=4.09A
Iocp*DCR=(Rocset*9.5uA)=(15.95*1.3*10m; Roset=21.8K
now chose Roset=22K
Csen=L/(DCR*Roset)=1uH/(10m*22k); Csen=0.00523uF
now chose Csen=5600pF
Iocp_min=(22K*9.5uA)/(10m ohm*1.3) =16.07A
Iocp_max=(22K*10.5uA/(10m ohm )=23.1A
```




```
DCR 10m ohm(max)   Cout ESR=15m ohm
1.8VP( Ibudget=9.91A )   Ipeak=9A, Imax=6.3 A
Fsw=1/1.5E-10*18.2k =366K
Vo=Vref*((PR97+PR99)/PR97)
Iocp=9*1.2=10.8A
Iocp*DCR=(Rocset*9.5uA)=10.67*1.3*10m; Roset=14.78K
now chose Roset=14.7K
Csen=L/(DCR*Roset)=1uH/(10m*14.7k); Csen=8.16nF
now chose Csen=8200pF
Iocp_min=(14.7K*9.5uA)/(10m ohm*1.3) =10.74A
Iocp_max=(14.7*10.5uA)/(10m ohm )=15.43A
```

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Issued Date	2007/09/20	Deciphered Date	2008/09/20	Title: 1.8VP / 1.05VSP
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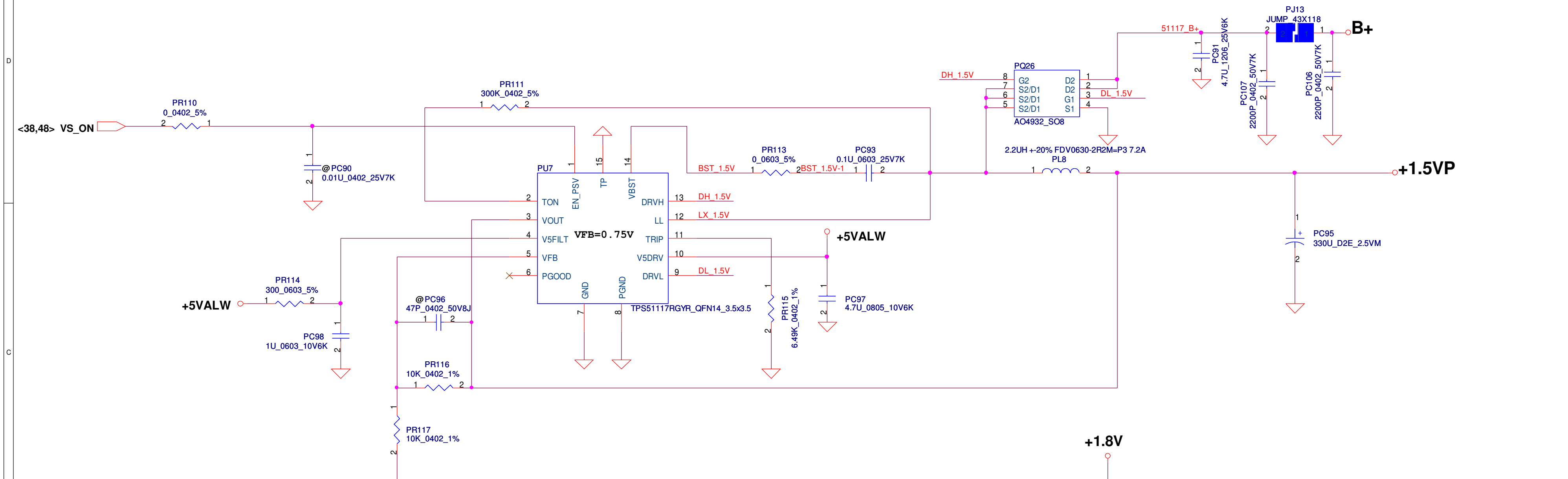
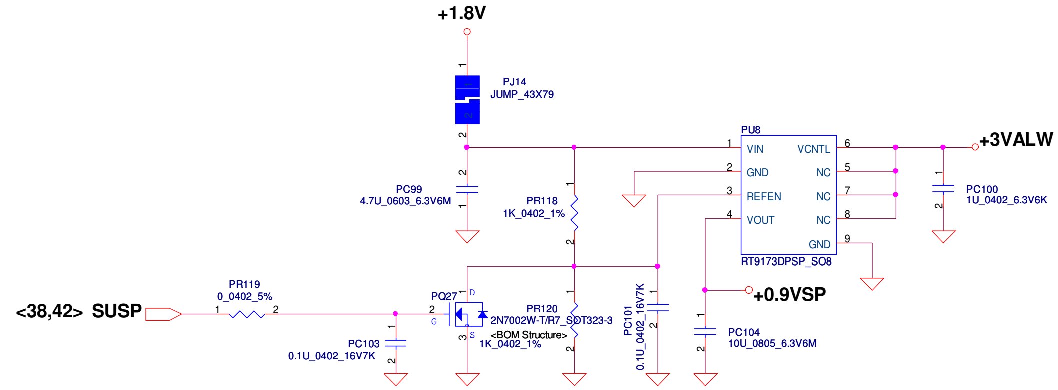
				Date: Monday, April 27, 2009; Sheet 48 of 53

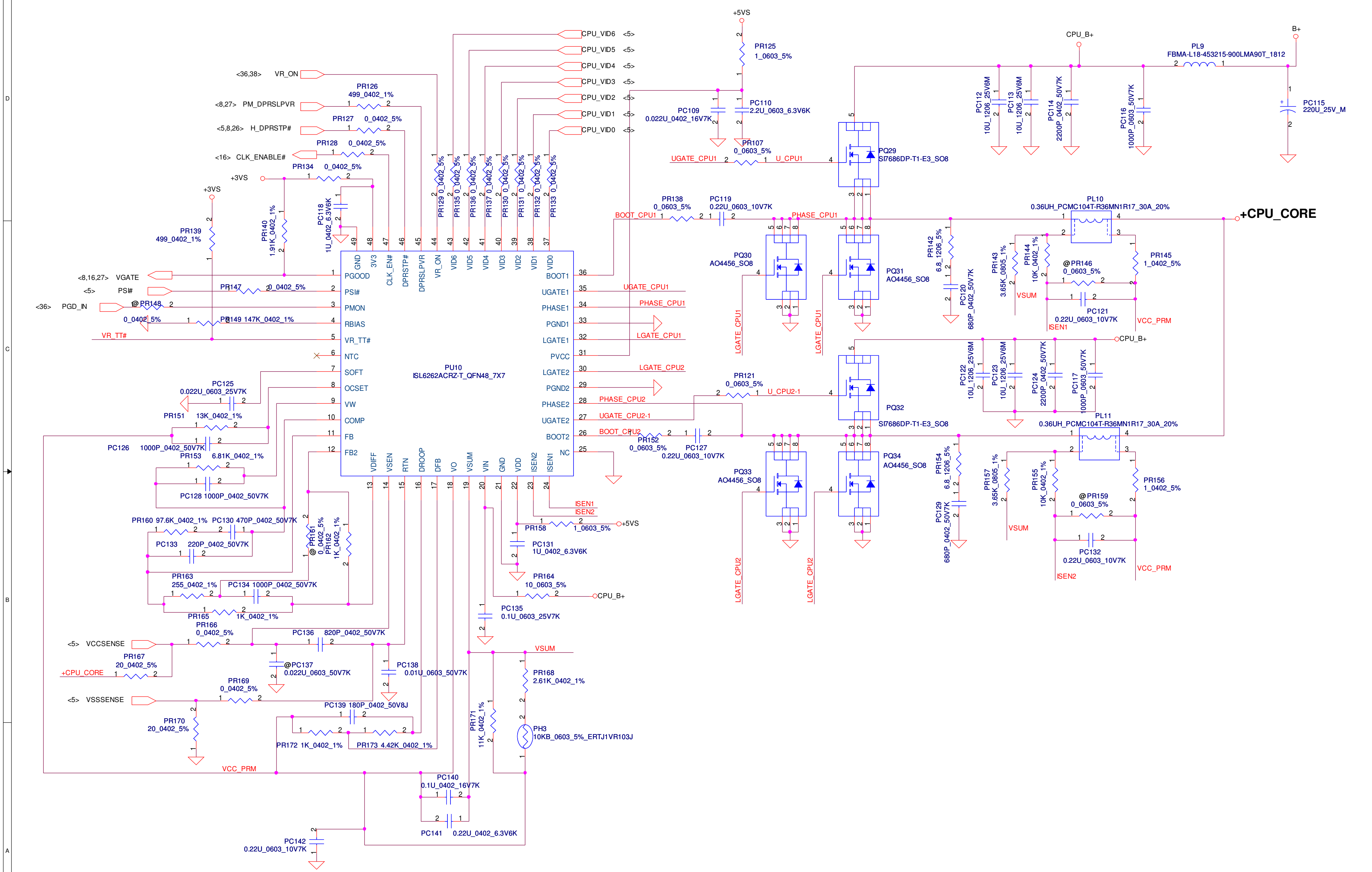


VFB=0.75V
Vo=VFB*(1+PR116/PR117)=0.75*(1+10K/10K)=1.5V
Ton=300K
Fsw=262KHz

Cout ESR=15m ohm (Ibudget=3.46A)
MOS Rds(on)=19.6 m ohm
Ipeak=2.84A, Imax=2A Iocp=3.48A
Delta I=((19-1.5)*(1.5/19))/(L*Fsw)=2.4A
=>1/2DeltaI=1.2A
Vtrip=Rtrip*10uA=6.49K*10uA=0.0649V
Iocpmin=Vtrip/Rdsonmax*1.2+1.46A
=0.0649/(0.0196*1.2)+1.2=2.76A+1.2A=3.96A
Iocpmax=(0.0649/(0.016))+1.2A=5.26A
Iocp=3.96~5.26A



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2007/09/20	Deciphered Date	2008/09/20	Title: +1.5VP/+0.9VSP
				Size: Custom; Document Number: KALG0-; Rev: 0.1
				Date: Monday, April 27, 2009; Sheet 49 of 53



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2007/09/20	Deciphered Date	2008/09/20	Title: +CPU_CORE
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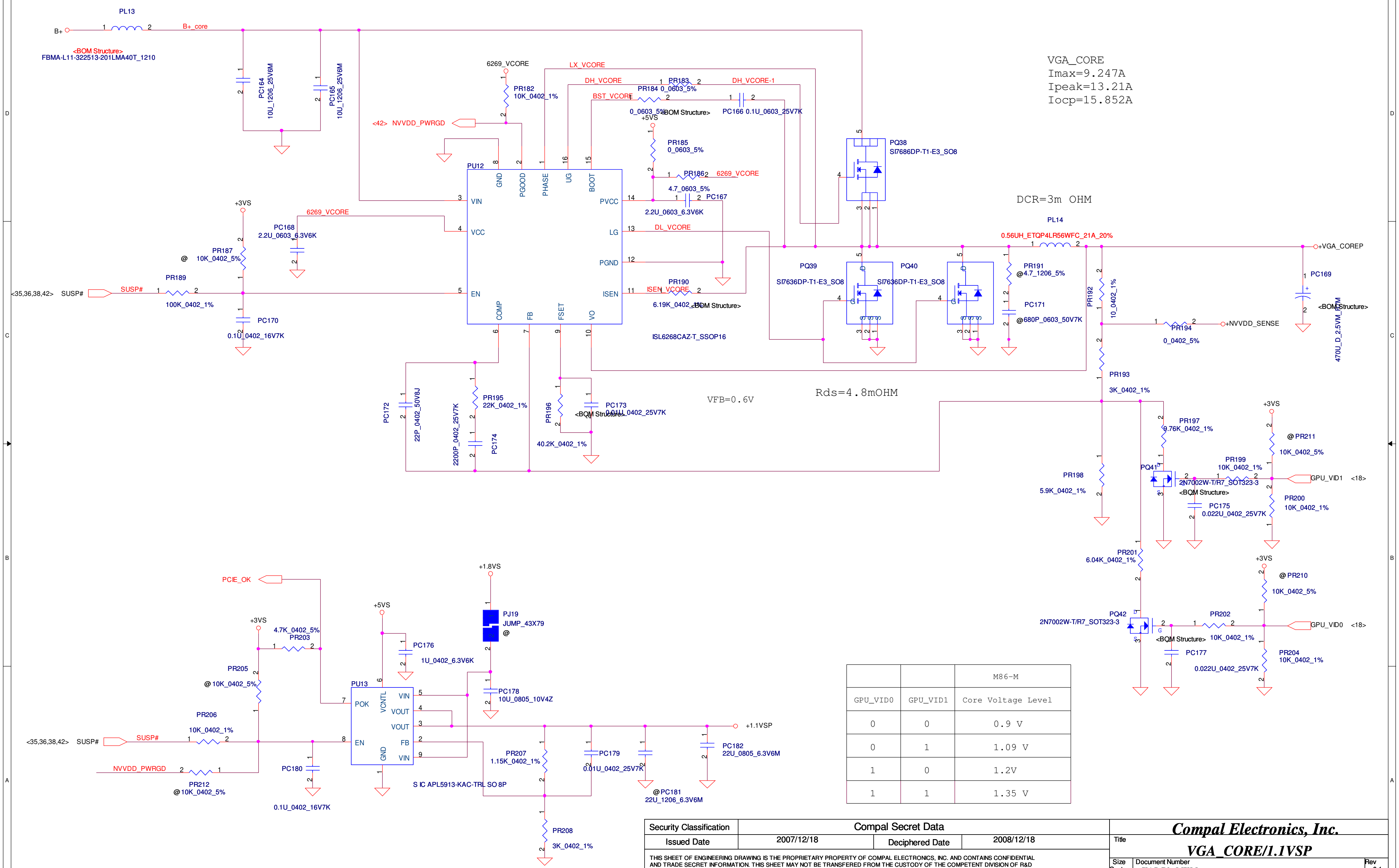
				Date: Monday, April 27, 2009; Sheet 50 of 53



		M86-M
GPU_VID0	GPU_VID1	Core Voltage Level
0	0	0.9 V
0	1	1.09 V
1	0	1.2V
1	1	1.35 V

Security Classification	Compal Secret Data	
Issued Date	2007/12/18	Deciphered Date 2008/12/18

Compal Electronics, Inc.

Title: **VGA_CORE/1.1VSP**

Size: Custom | Document Number: KALG0- MXM | Rev 0.1

Version change list (P.I.R. List)

Item	Fixed Issue	Reason for change	Rev.	PG#	Modify List	Date	Phase
1	Change PR196 value	VGA OCP	0.3	50	change the resistance value of pr196 from 57.6K to 40.2K	2009/02/06	PVT
2	Change PL14 value	prevent OVP occur	0.3	50	change the inductance value of pL14 from 1uH to 0.56 uH	2009/02/06	PVT
3	Change PU4 IC part number	vender suggestion	0.3	45	change part number to SA00002V400	2009/02/06	PVT
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							

B --> C Change List

0220-------------------
Page 6, Change C98 BOM structure to @

0218---------------
Page 20, Add R569 with BOM structure @
Page 23, R34, R38, R40, R53, R54, R55 change BOM structure to @
Page 31, Change R215 to 18K
Update Power Schematics

0213-----------------------------------
Page 11, Delete R113
Add J1 JUMP_43X79 with BOM structure @
Page 31, R558, R559 0 ohm with BOM structure @
R560, R561 4.7K with BOM structure @
Page 32, R562, R563 0 ohm with BOM Structure @
R564, R565 0 ohm
Page 38, Delete C619
Add R566, R567, R568 0_0603 with BOM Structure JAL90@
0204---------------------------------
Page 12, Change L31 to MBK1608301YZF_0603 with BOM structure GM@
Change R163 to 0_0805_5%
Page 23, Change BOM Structure of U5 to @
Page 27, Change BOM Structure of R555 and R550 to @
Page 33, Change R217 to 31.6K_0402_1%
Change C313 to 1U_0603_10V6K
Page 34, Change R503 to FBMA-L10-160808-301LMT_0603
01/31---------------------------------
Page 23, Delete U10
01/29---------------------------------
Page 23, Change R152 BOM Structure to @
01/24--------------------------------
Page 4, Change U8 to SA00001Z700 (EMC1402)
Page 33, Change C338 to SE076104K80
Page 35, Mount C584
01/23---------------------------------
Page 38 Delete F3, R558~560, C609~614
01/22-------------------------------------
Page 11, Delete R79
Change J1 Symbol to JUMP_43X79
Page 33, Add R557 10K (Check)
Change R245 BOM Structure with @
Page 38, Add C609~614, R558~560 (Check)
C607,608, 615~619 (Check)
01/17-------------------------------------
Page 11, Add R79 0_0805
Update Power Schematics
01/16------------------------------------
Page 11, Delete R79 0_0805
Add J1 JUMP_43X79
Page 16, Change C296, R301 to 27P_0402
Page 19, Change L17, L19, L21 BOM structure to GM@
Page 23, Mount U29, R339
Add U10 with BOM structure @ (Co-lay with U5)
Change R340 Bom structure to @
Change U5 to MX25L4005AMC-12G_SO8 (SA00002A900)
Page 27, Change U26, C420 BOM structure to @
Change R550 to 0_0402
Add R555 0_0402
Page 32, Change R269 to 240_0402_5%, R267 to 453_0402_1%
Change R268 pin1 connect to +5VALW
Page 33, Change R217 to 18K_0402_1% with BOM structure PM@
Page 35, Add R551,R552, R553, R554 75_0603_1% with BOM structure JAL90@
Add D32 PJDLC05_SOT23-3
Page 38, Add F3 3A_15VDC_SMD2920P300TF/15
Page 49, Add R551,R552, R553, R554 1K_0603_1% with BOM structure 268@
Add L17, L19, L21 0_0805 with BOM structure PM@
Update U38 (ALC268-VB1-GR) PN:SA00001GD10 for JAW50

Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2008/03/28	Deciphered Date	2008/09/20	Title: HW PIR1
				Size: Custom; Document Number: KAL90; Rev: 0.1
				Date: Monday, April 27, 2009; Sheet 53 of 53